



sAlfer Lab

Joint lab on Safety and Security of AI

Computer Vision Technologies and Biometrics Fingerprints

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**M.Sc. Degree in Computer Engineering, CyberSecurity and
Artificial Intelligence**



University
of Cagliari, Italy

Department of Electrical
and Electronic
Engineering



Outline of the talk



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Introduction to biometrics



Fingerprints

What are they
Pre-processing
Feature extraction
Matching



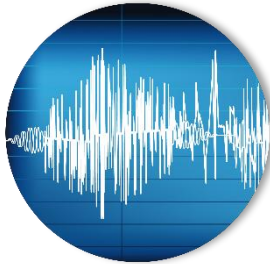
Presentation Attack Detection

Modules
Design
Features
Classifier
Deep learning

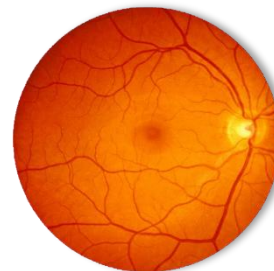
Definition



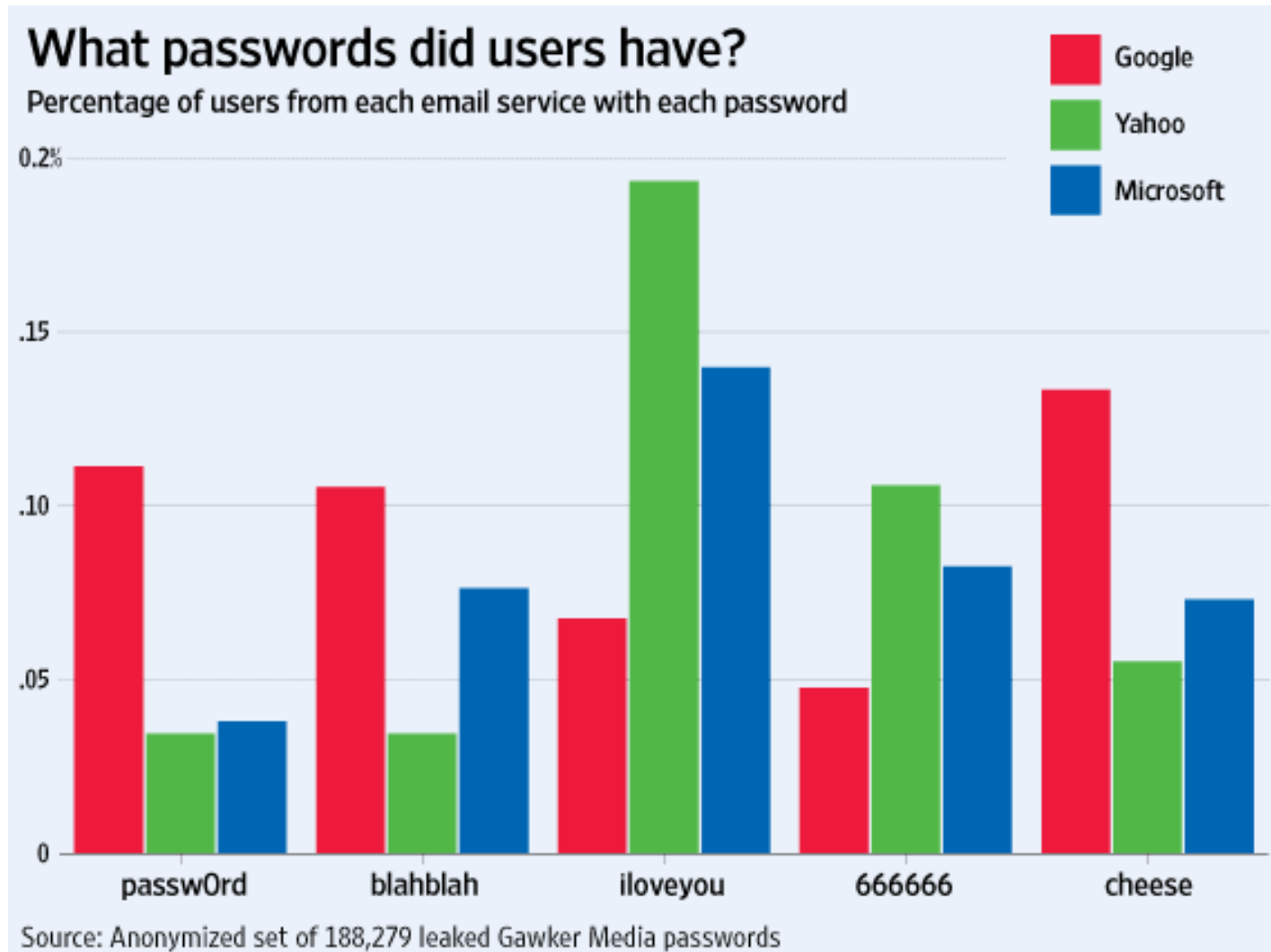
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Something you are



Something that *only* you should know



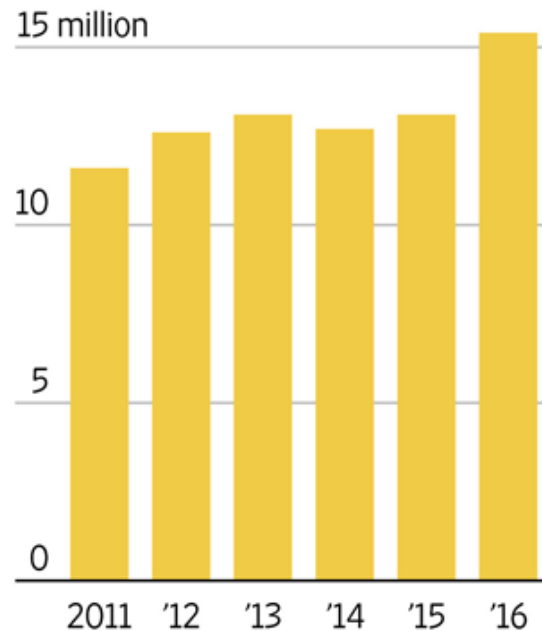
Identity theft and cybersecurity



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Card Fraud

U.S. adult victims of identity fraud, in millions

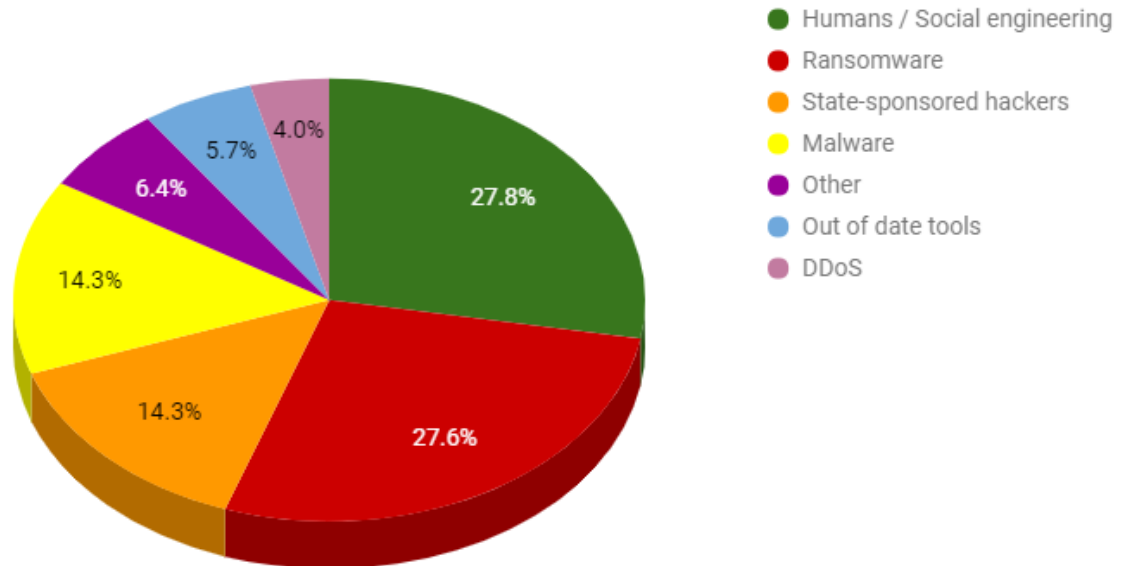


Note: Includes fraud in credit and debit cards, bank accounts, email payments and mobile phone accounts.

Source: Javelin Strategy & Research

THE WALL STREET JOURNAL.

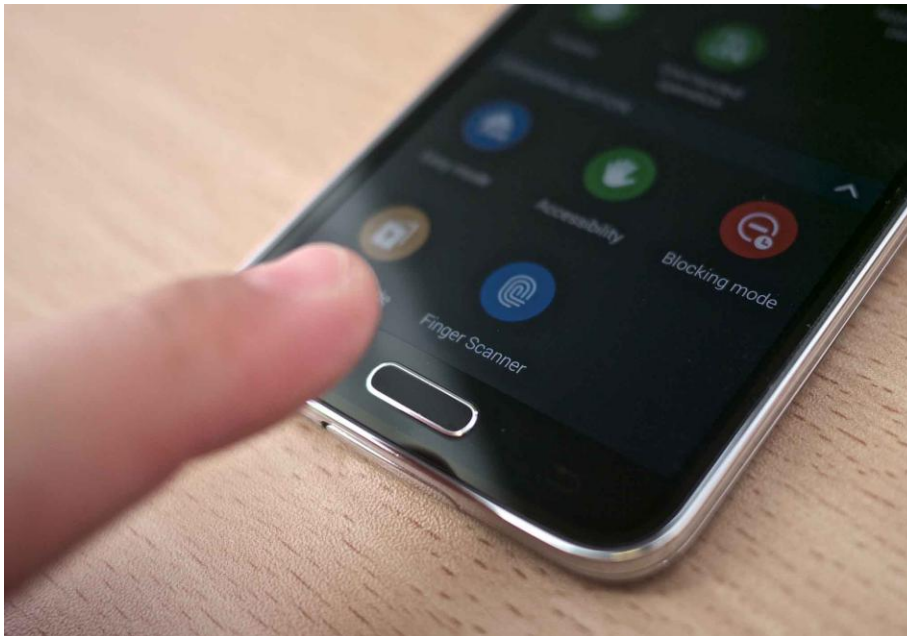
What is your biggest cybersecurity concern?



- Humans / Social engineering
- Ransomware
- State-sponsored hackers
- Malware
- Other
- Out of date tools
- DDoS

Biometric systems and «security chain»

- Identity check is the first stage in current «security chain»



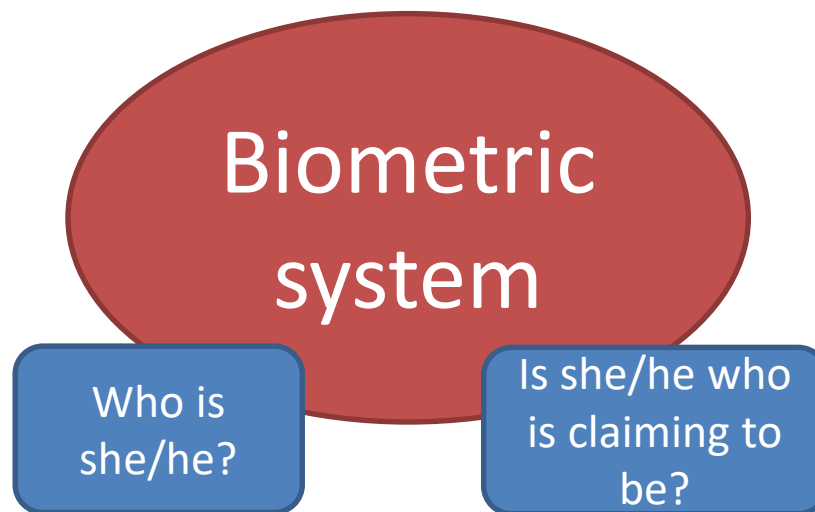
Biometric properties



	Universality	Uniqueness	Permanence	Collectability	Performance	Acceptability	Circumvention	Cost	Maturity
Fingerprint	M	H	H	M	H	M	L	M	H
Face	H	L	M	H	L	H	H	L	M
Iris	H	H	H	M	H	L	L	H	M
Hand vein	M	M	M	M	M	M	H	H	L
Facial thermogram	H	H	L	H	M	H	H	H	M
Retinal scan	H	H	M	L	H	L	H	H	H
Ear	M	M	H	M	M	H	M	M	L
Hand geometry	M	M	M	H	M	M	M	L	H
Voice	M	L	L	M	L	H	H	L	H
Signature	L	L	L	H	L	H	L	M	M
Gait	M	L	L	H	L	H	M	H	L



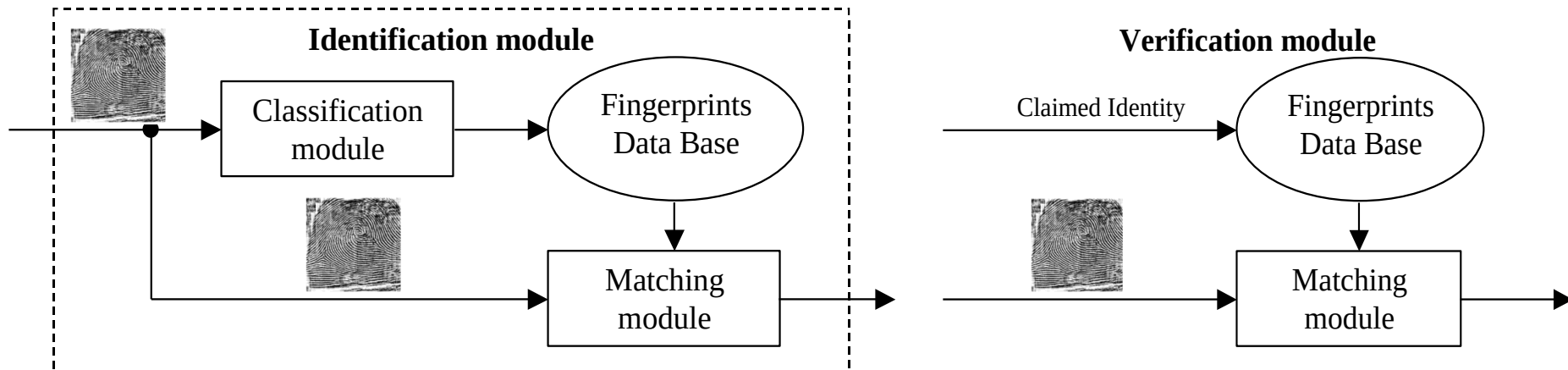
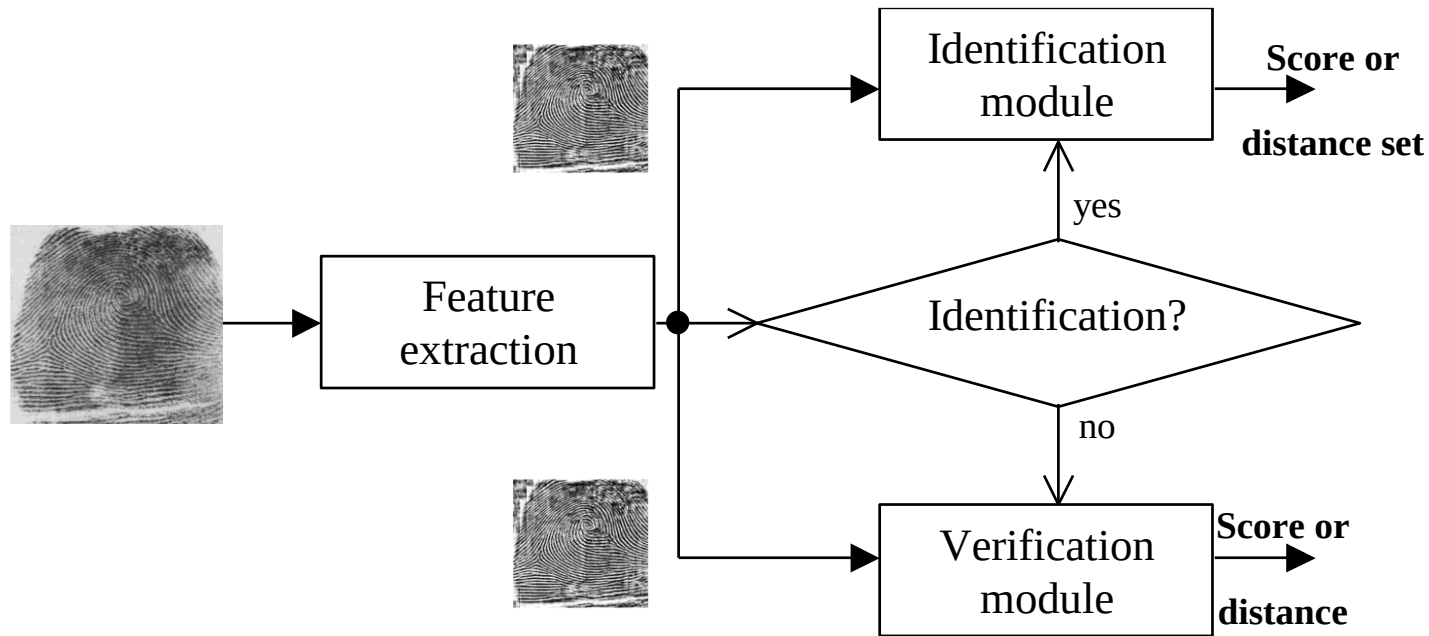
- **Biometric systems are pattern recognition-based algorithms**
- **They require that the system is «trained»**
 - By collecting a set of the target users «templates»
 - By embedding the uniqueness measurements of users as system's parameters



Biometric system's rough view



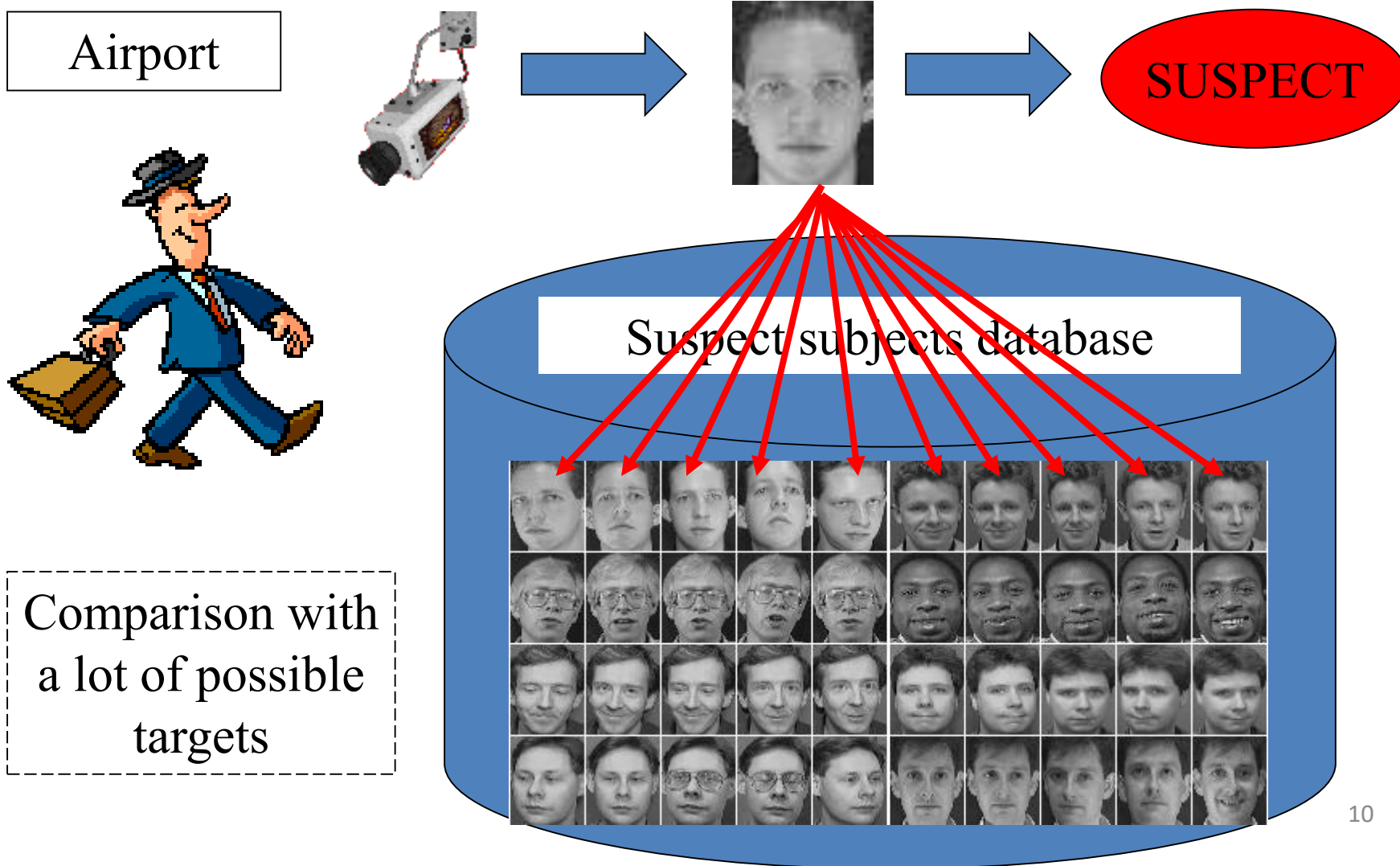
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Personal recognition...



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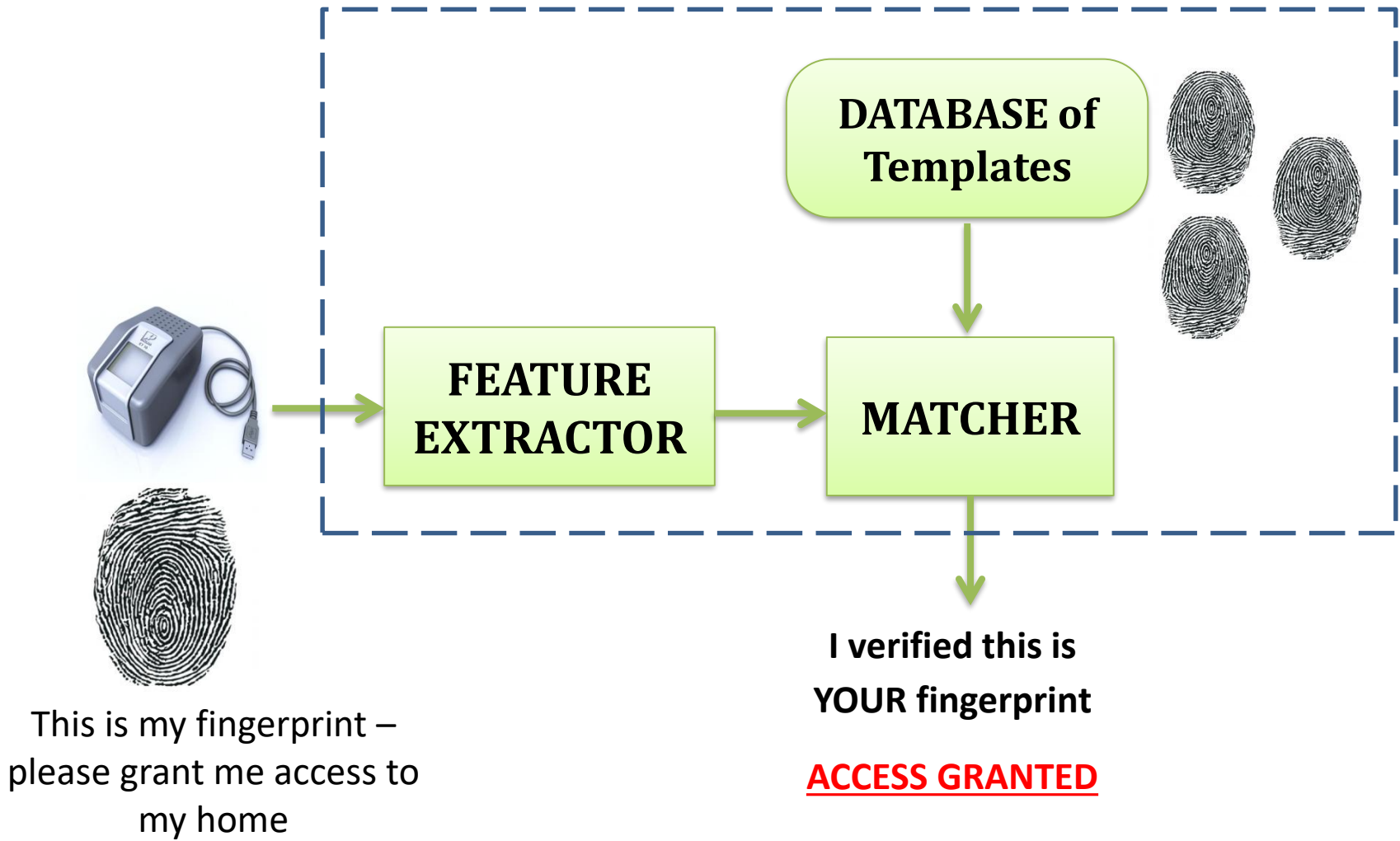


10

...and personal verification



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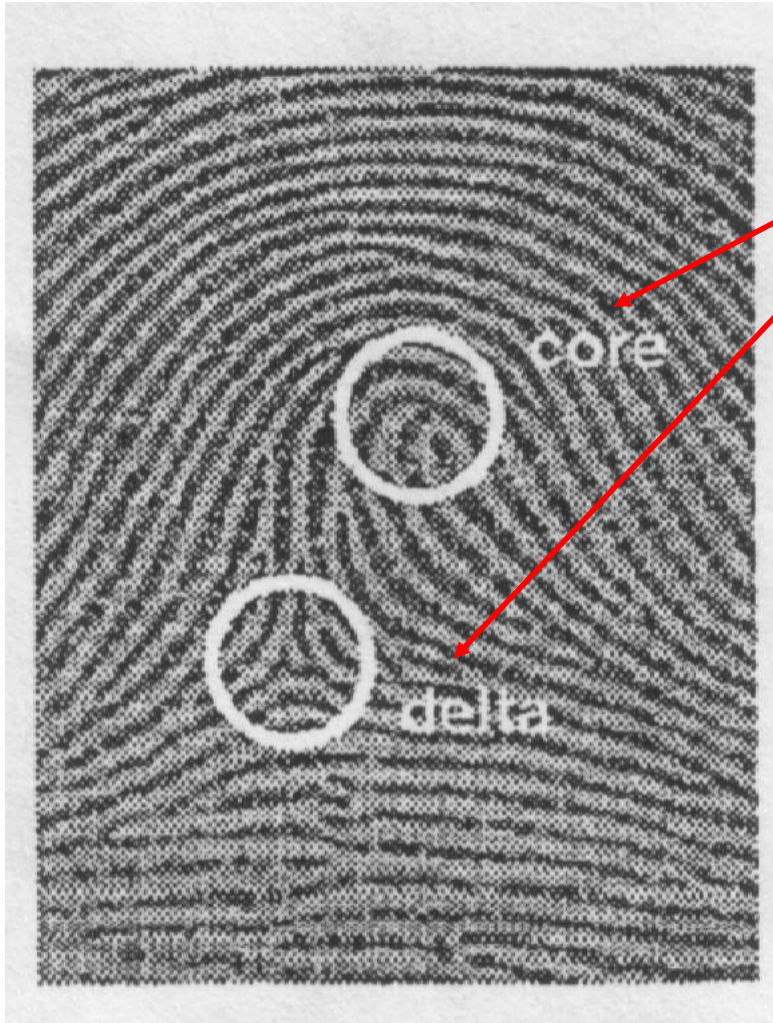
Let's go in depth with...



Characteristics of fingerprints



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Singular points

Minutiae



Ridge termination



Ridge bifurcation

Fingerprint classes



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Whorl



Right Loop



Left Loop



Arch

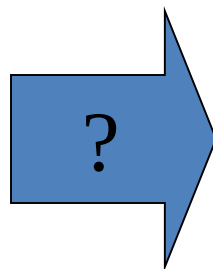


Tented Arch

Cross-referenced fingerprints



A



T



A & T

Uniqueness of fingerprints



It is widely acknowledged that fingerprints are unique from person to person



However, no scientific proof has been provided to support this claim



This conviction derives from the wide variety of fingerprints in terms of:

Intra-person variations

Extra-person variations



Some statistical analysis has been done on the basis of bernoullian model, using minutiae as features

Statistical evaluation of uniqueness of fingerprints



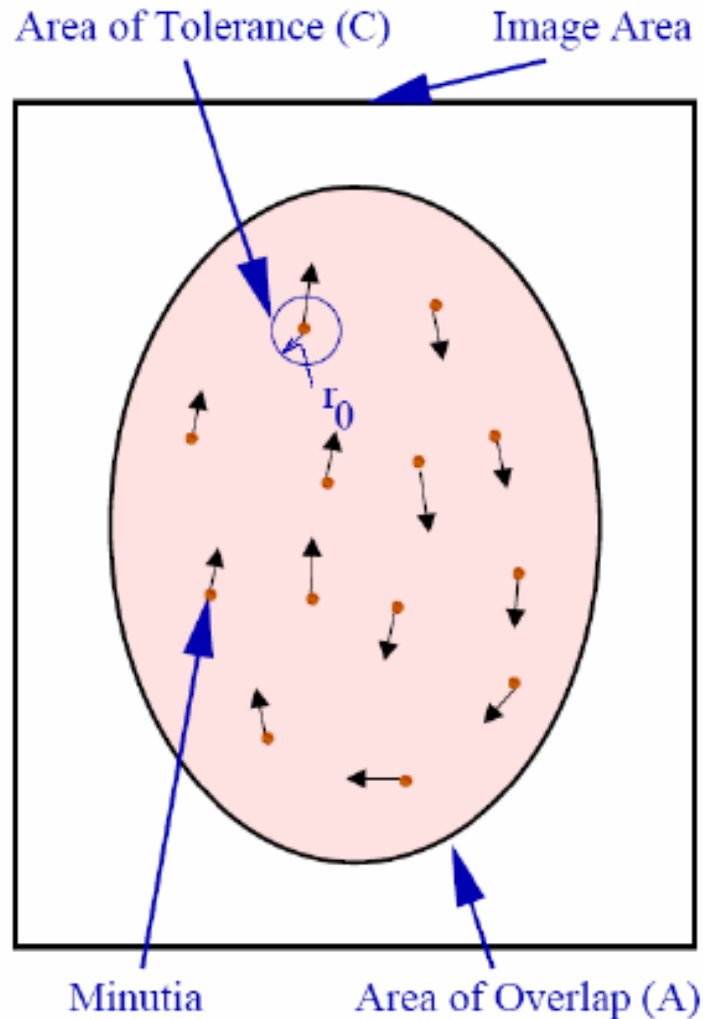
The probability of finding more than q common minutiae in two fingerprint images having m and n minutiae, respectively, can be modelled as follows:

$$p(M, m, n, q) = \sum_{\rho=q}^{\min(m,n)} \left(\frac{\binom{m}{\rho} \binom{M-m}{n-\rho}}{\binom{M}{n}} \times \binom{\rho}{q} (l)^q (1-l)^{\rho-q} \right)$$

Hyper Geometric Binomial



Some numbers...



(a) $M=52$
 $m=n=q=26$

$$P = 2.40 \times 10^{-30}$$

(b) $M=52$
 $m=n=26, q=10$

$$P = 5.49 \times 10^{-4}$$

$$M = A/C$$

Homework (verification)



$$\frac{\binom{m}{\rho} \binom{M-m}{n-\rho}}{\binom{M}{n}}$$

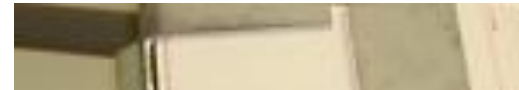
- M is the total number of minutiae
 - m is the number of minutiae in the ROI
 - n is the number of observed minutiae in the latent print
 - ρ is the number of minutiae for which you want to compute the probability of match
- **Let's assume:**
 - Each fingerprint contains **20 minutiae**
 - You collect a **partial fingerprint** from a crime scene, containing only **8 minutiae**
 - You want to know the probability that ρ of those 8 minutiae match a stored fingerprint of the suspect

Fingerprint sensing

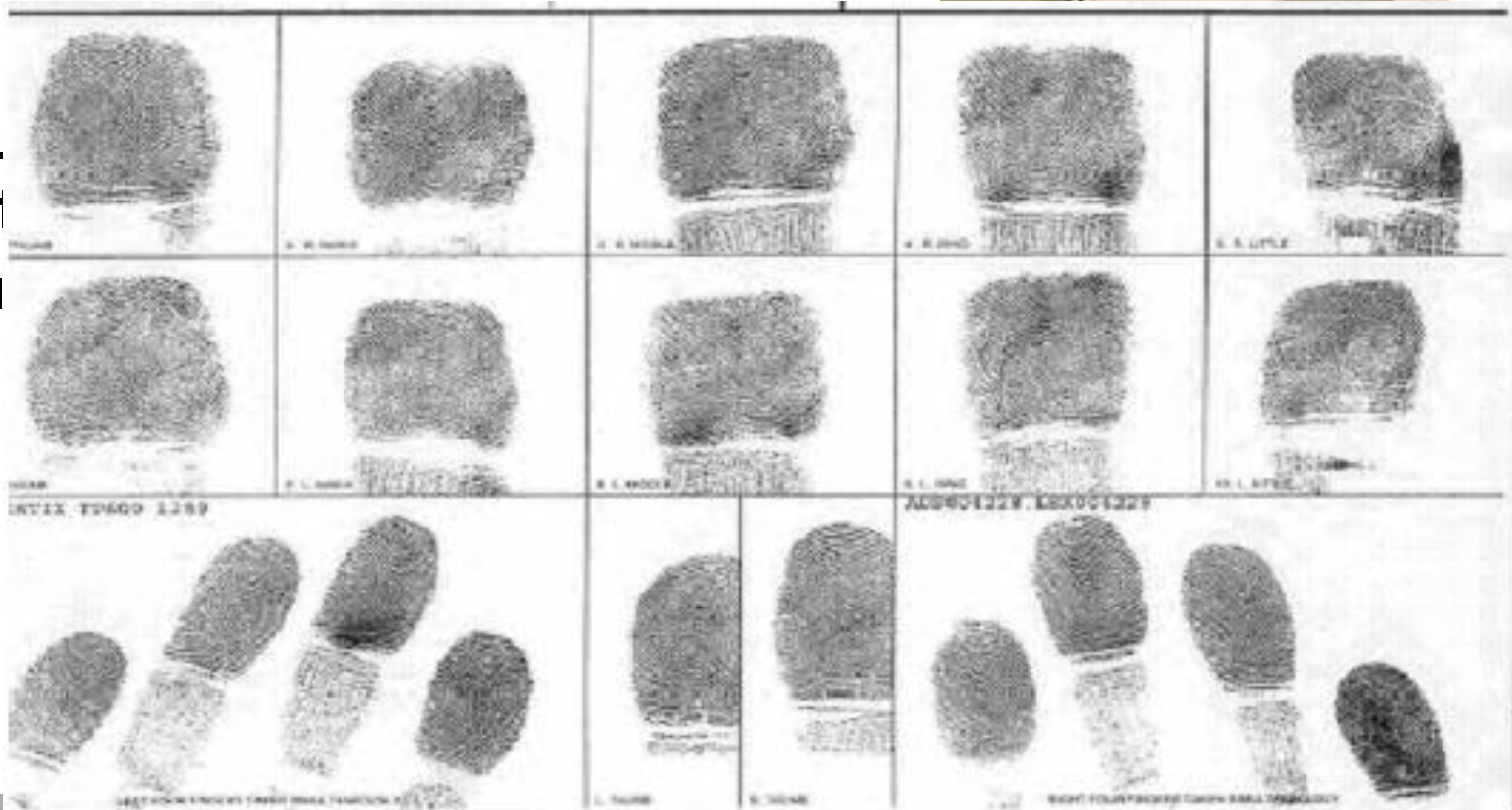


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- Ink-rolled



But
of
Fin



Fingerprint sensing by electronic scanners



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Solid-state



Optical



Sensors in the Biometric Unit's Lab



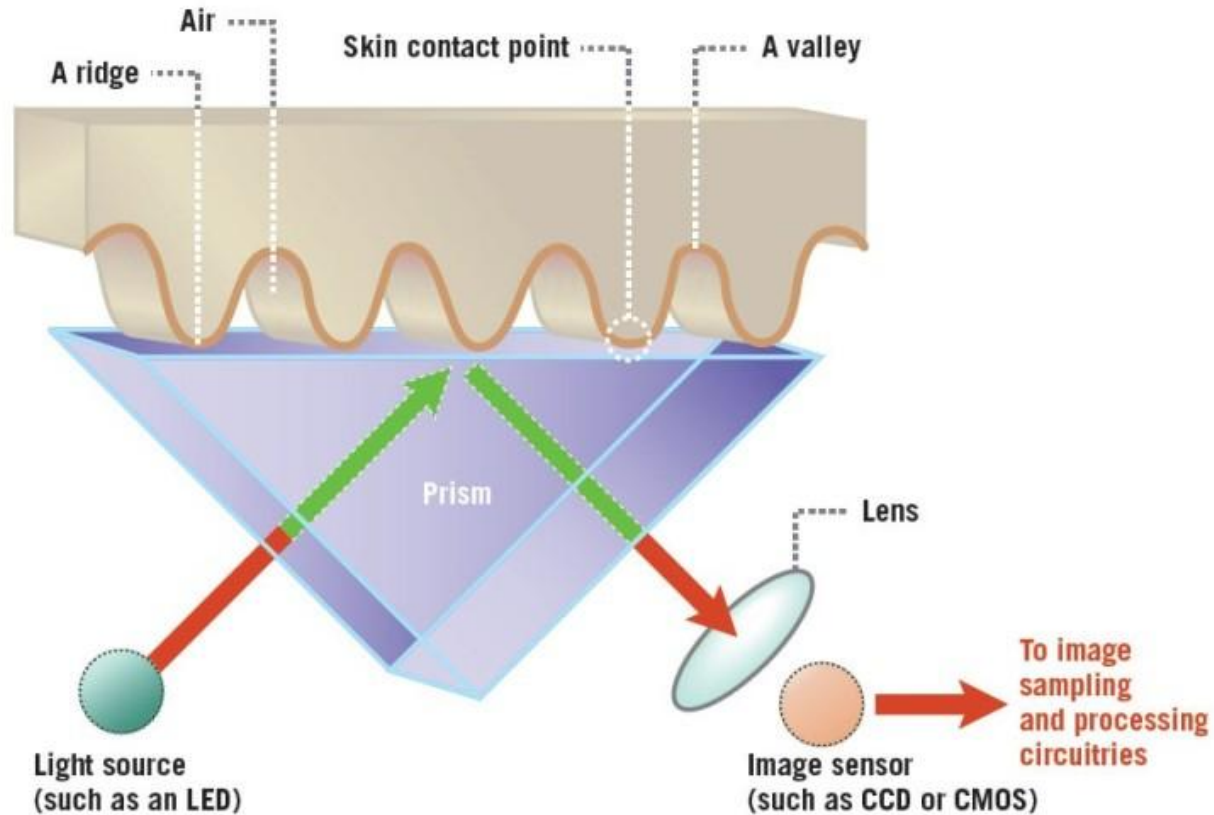
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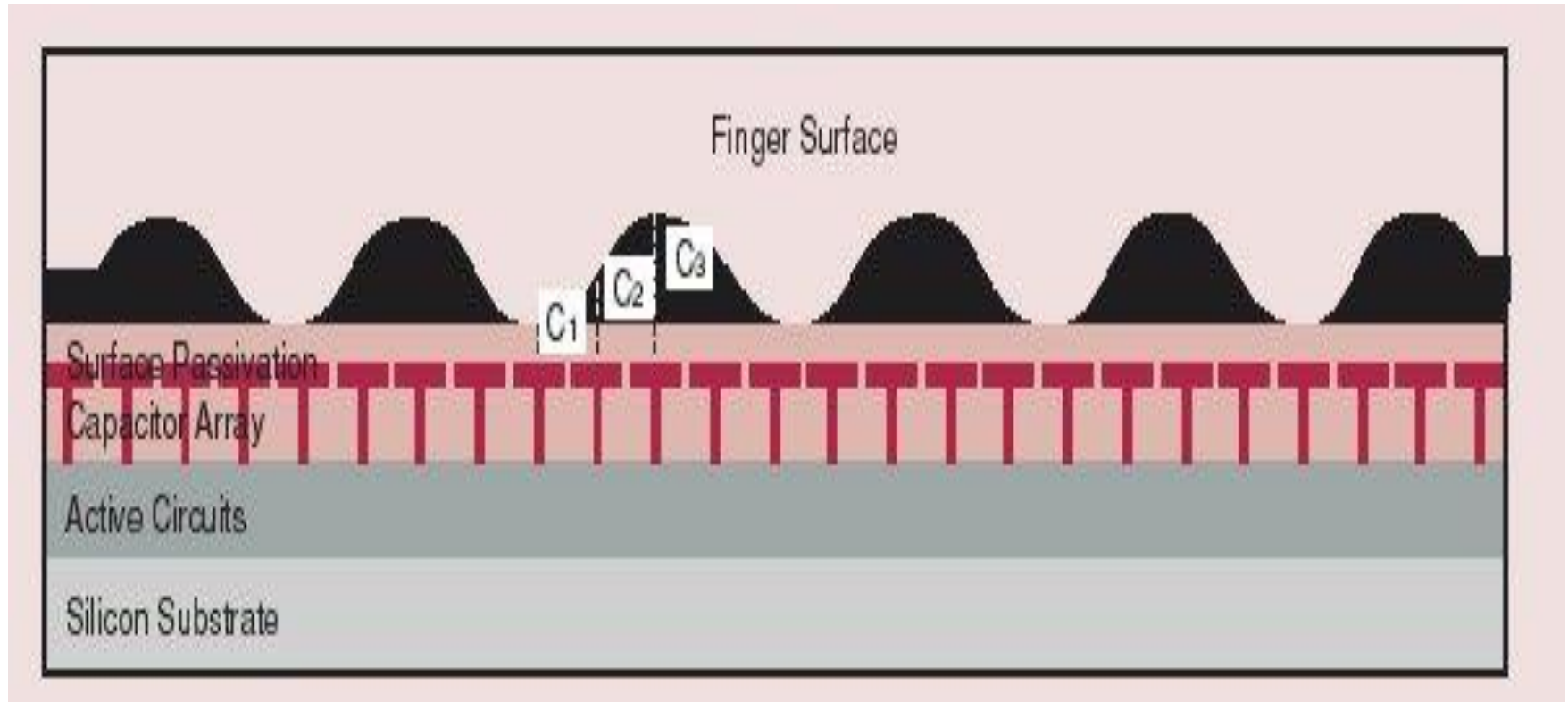
Optical fingerprint scanners



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Solid-state fingerprint scanners



Differences among optical and capacitive fingerprint images



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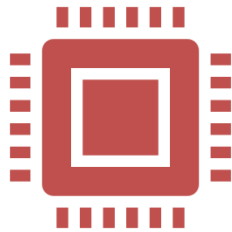
Optical Sensor



Bad pressing



Capacitive
Sensor



Optical sensors

Wide acquisition surface

Good quality acquisition

High resolution reached (500-1000 dpi)

Difficult to embed



Capacitive sensors

Small acquisition surface

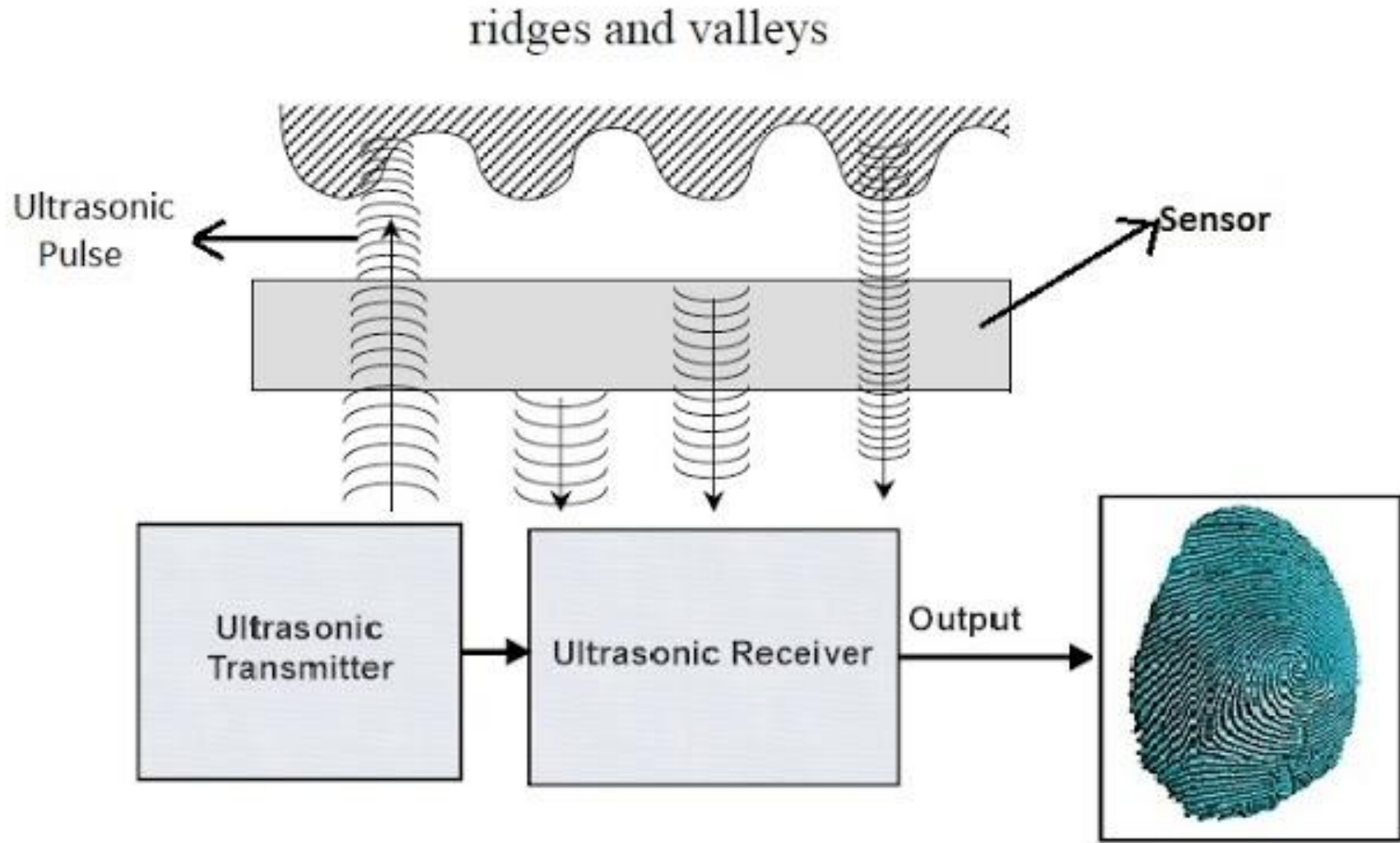
Cost increases with resolution (500 dpi)

Easy to embed

Ultrasonic sensors



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Contactless sensors



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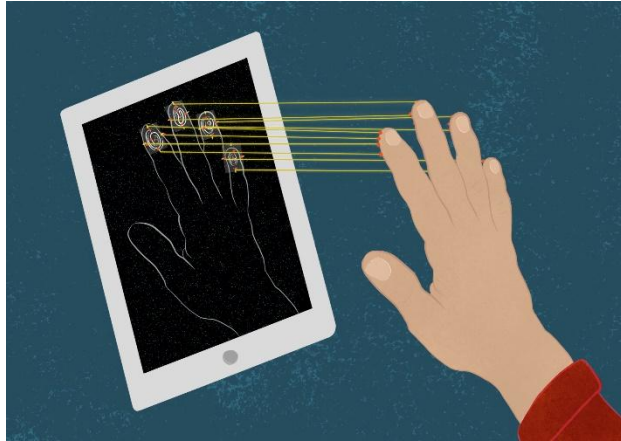
- Spectral signature is obtained by illuminating the skin
- Light is polarized in five different wavelengths



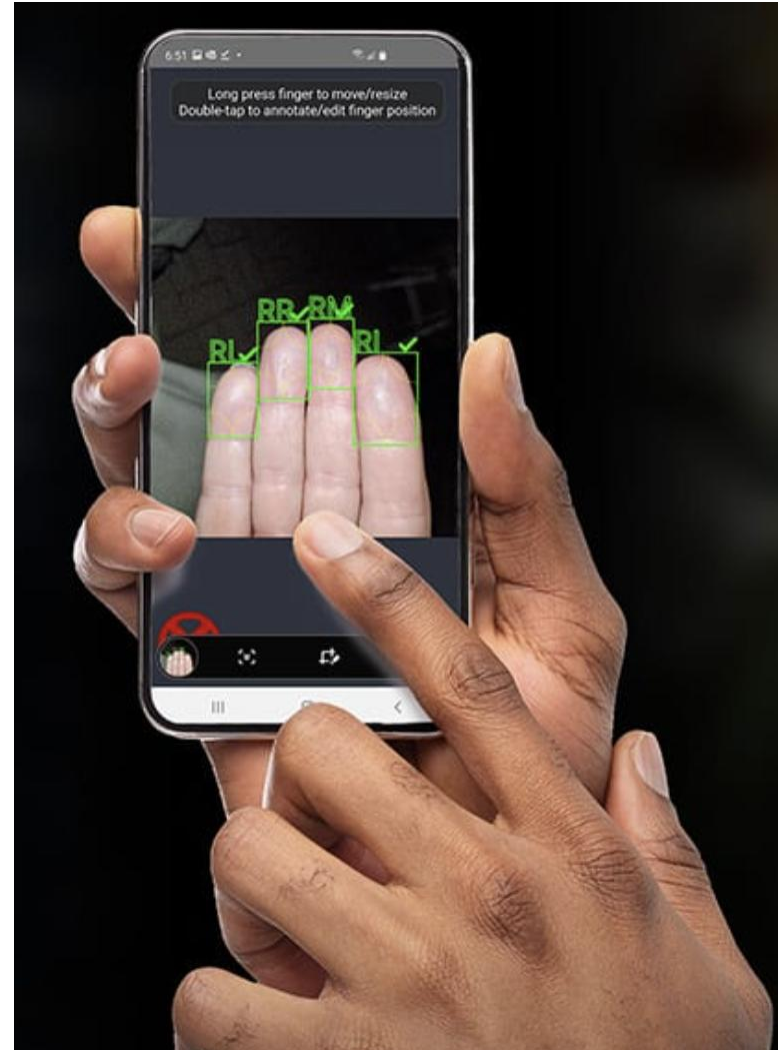
A challenging frontier



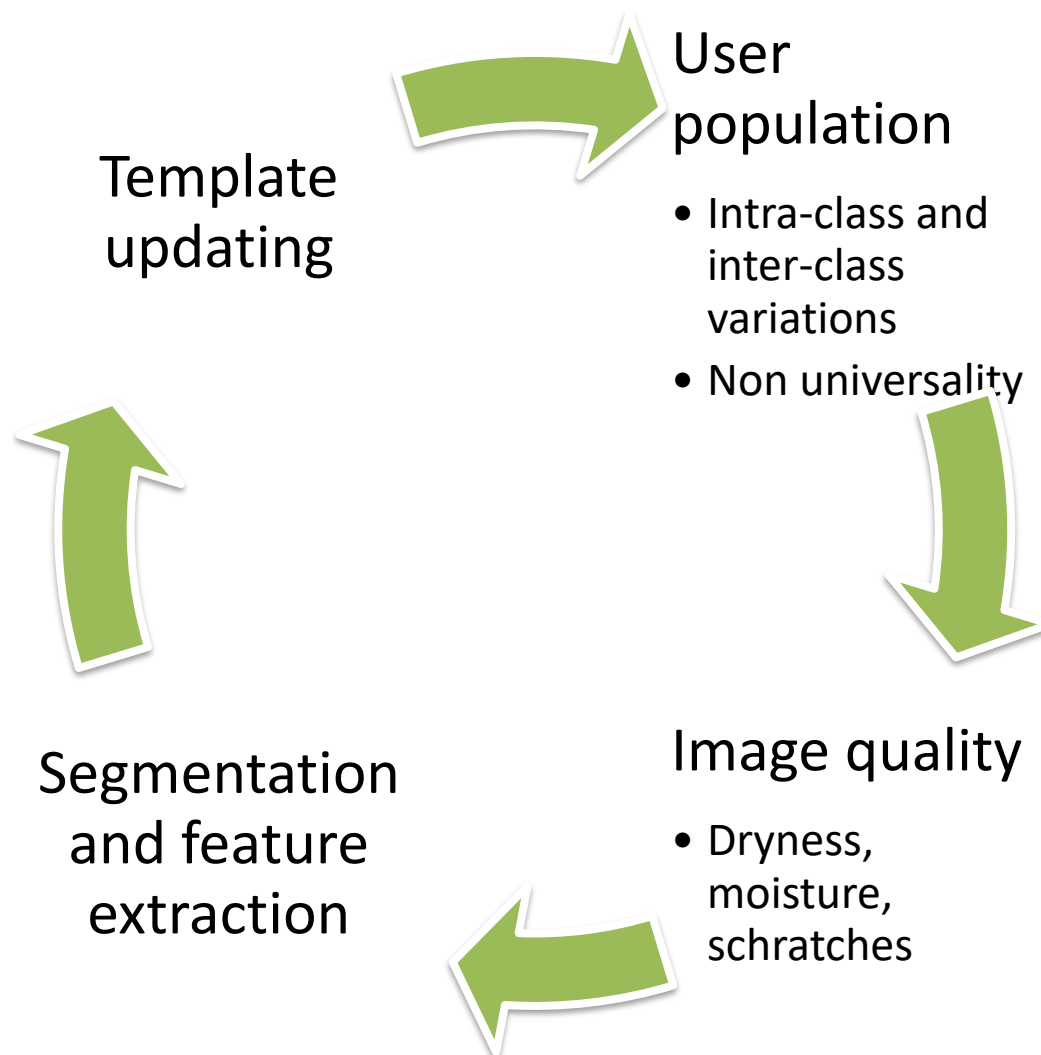
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- Resolution?
- Distance?
- Lighting?
- How many details?



Problems in fingerprint recognition



User population problems

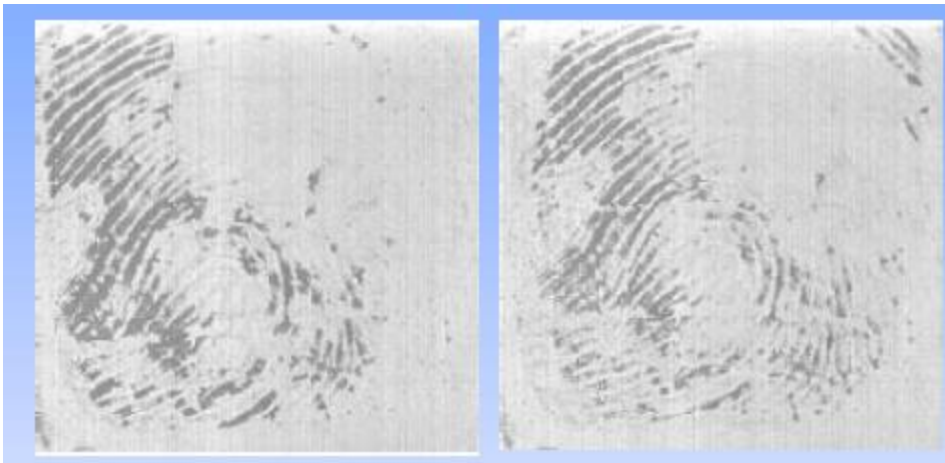


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Intra-class variations



Inter-class variations



Around 3% of user population has intrinsically poor quality fingerprint images

Rigid and elastic deformations



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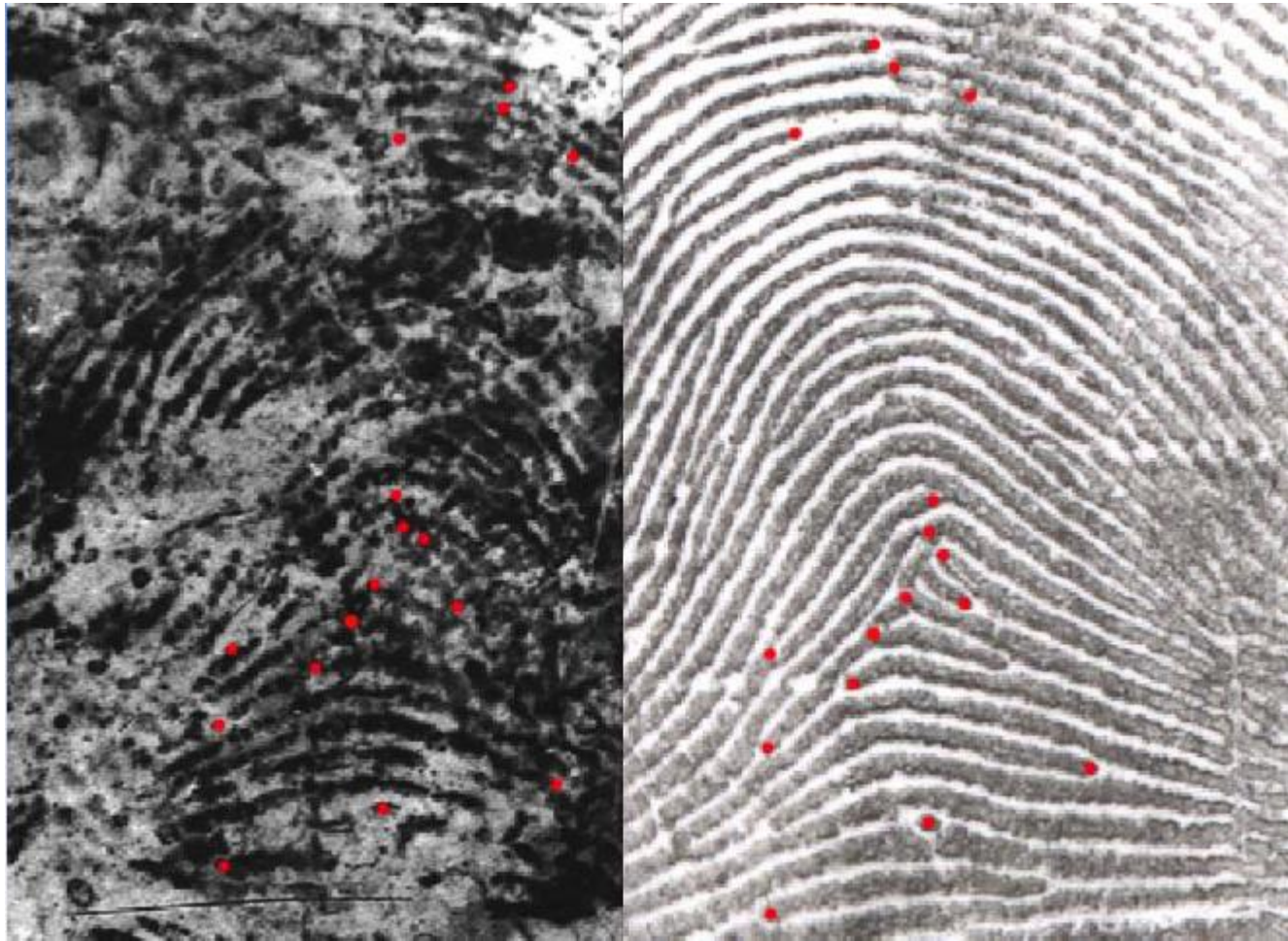


Nonlinear Rotation



Nonlinear lateral movement

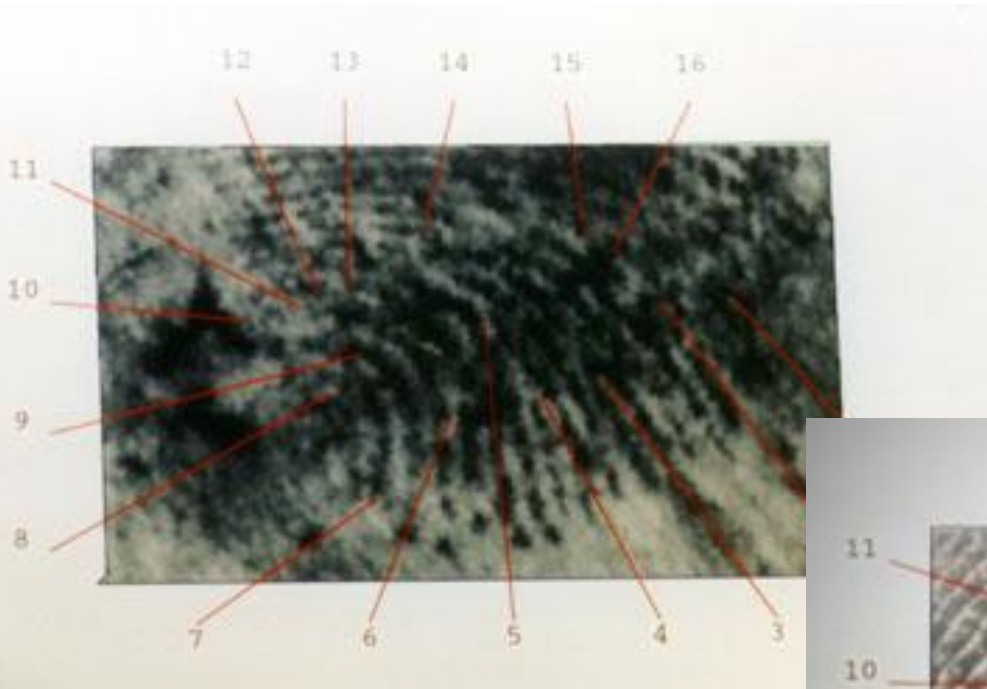
Matching errors due to image quality: the Brandon Mayfield's fingerprint (Spain, 2004)



Errors and fingerprints (the Shirley McKie's, Scotland, 1997)

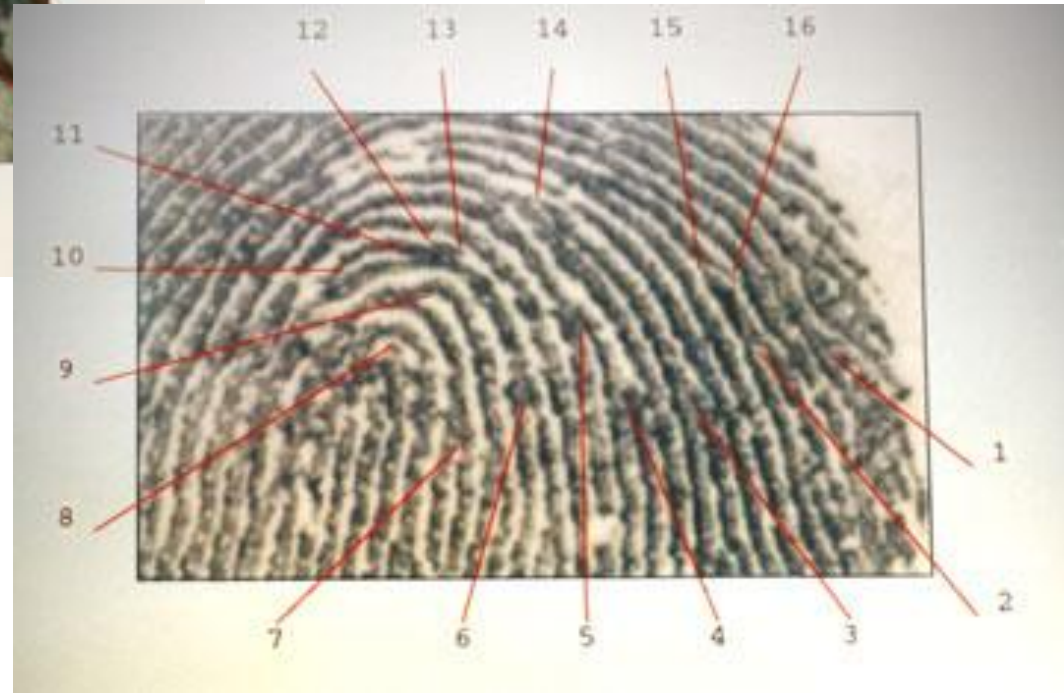


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Latent

Known



Fingerprint images processing steps

- Fingerprint enhancement
- Fingerprint classification
 - In 1:N classification
- Fingerprint matching
 - Feature extraction

Fingerprint enhancement

- It is aimed to improve the “quality” (visual quality) of fingerprint
- Ridges and valleys must be well-defined and separated as more as possible
- Main methods adopted:
 - 2D-Fourier transform
 - Gabor filters

- Image is tessellated in 32x32 sized blocks
- 2D FFT:
$$X_{jk} + iY_{jk} = \sum_{m=0}^{31} \sum_{n=0}^{31} (A_{mn} + iB_{mn}) \exp\left(\frac{-2\pi i}{32}(mj + nk)\right)$$
- Enh:
$$U_{jk} + iV_{jk} = \left(X_{jk}^2 + Y_{jk}^2\right)^{0.3} (X_{jk} + iY_{jk})$$
- Conv:
$$C_{mn} + iD_{mn} = \sum_{j=0}^{31} \sum_{k=0}^{31} (U_{jk} + iV_{jk}) \exp\left(\frac{2\pi i}{32}(jm + kn)\right)$$

Example



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- Designing a fingerprint enhancement module based on the FFT
- Evaluate differences by varying the power in the formula below

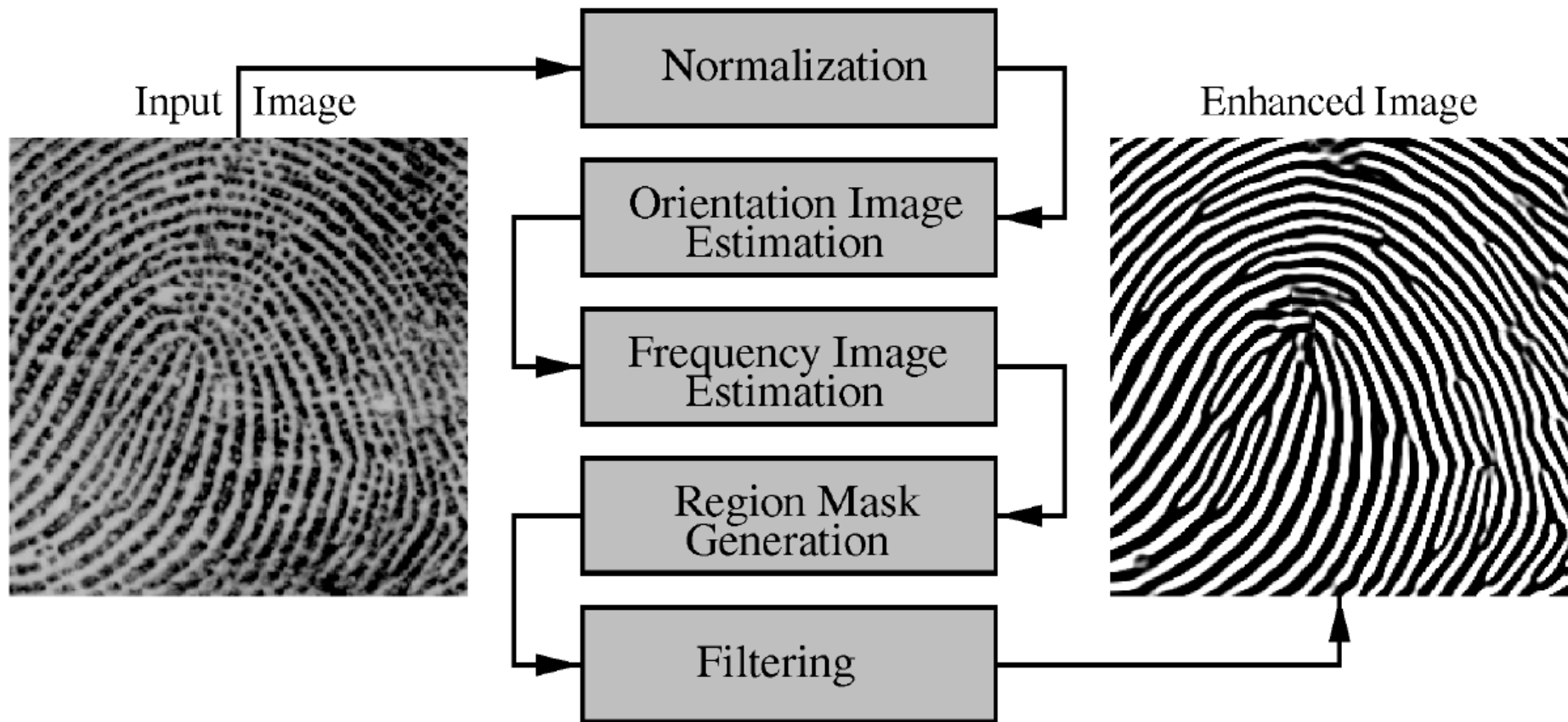
$$U_{jk} + iV_{jk} = (X_{jk}^2 + Y_{jk}^2)^{0.3} (X_{jk} + iY_{jk})$$

- Please send us your code using the Teams Activity – This activity will be soon set in the main page

Filter-bank based fingerprint enhancement



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Filtering is based on the Gabor wavelets transform

Orientation field



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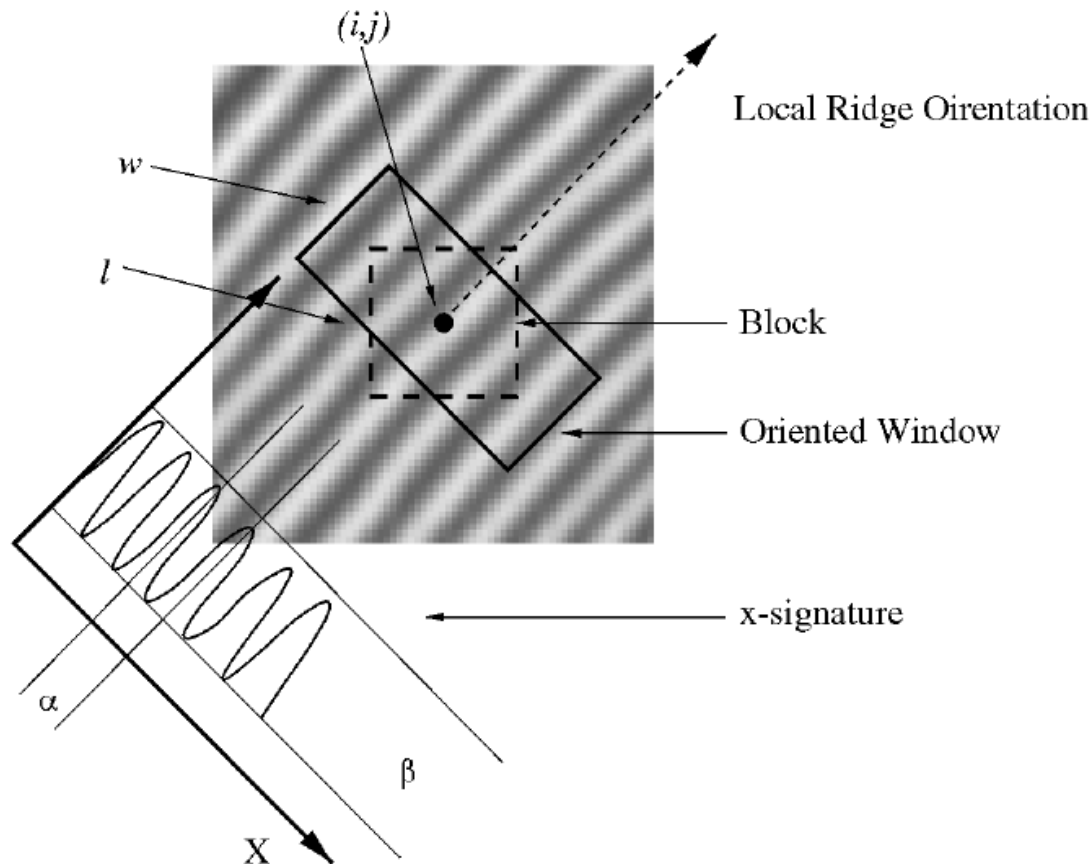
- Orientation of ridge flow along the image
- In general the basic idea is to estimate the derivative along x and y axes



Ridge frequency



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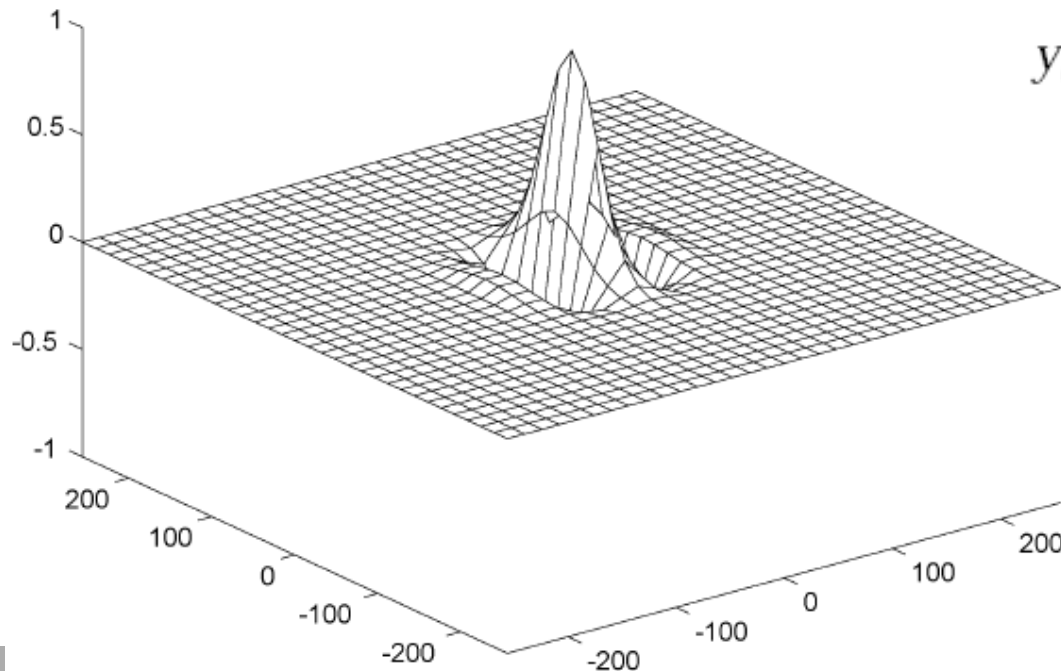
- Frequency of the alternance between ridge and valleys

- Gabor filter:

$$h(x, y; \phi, f) = \exp\left\{-\frac{1}{2}\left[\frac{x_\phi^2}{\delta_x^2} + \frac{y_\phi^2}{\delta_y^2}\right]\right\} \cos(2\pi f x_\phi),$$

$$x_\phi = x \cos \phi + y \sin \phi,$$

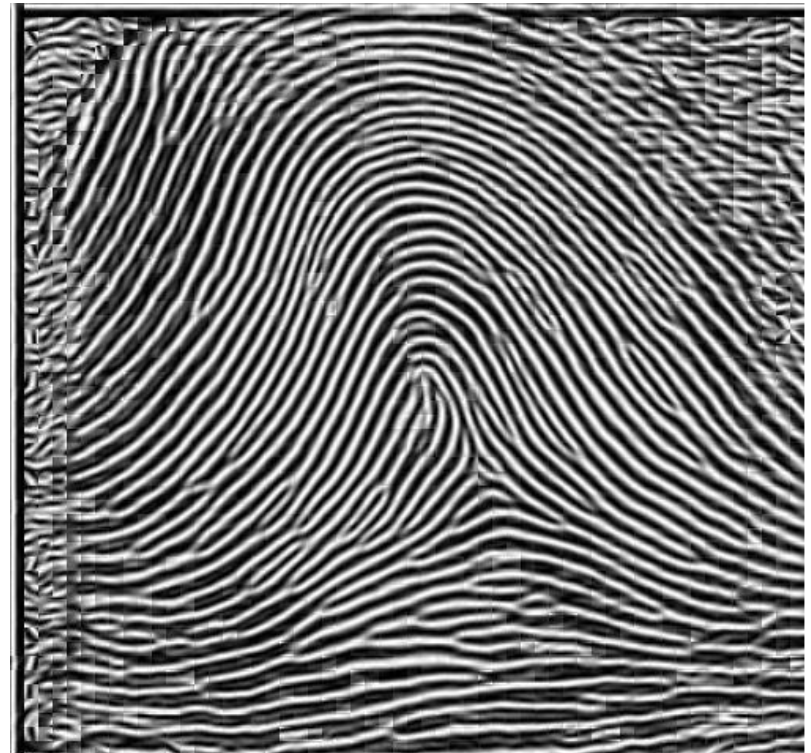
$$y_\phi = -x \sin \phi + y \cos \phi,$$



Example



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Tassellate the fingerprint image by varying the tassellation sizes



Compute the orientation field of each patch



Define a set of Gabor filters based on different orientations



Apply the Gabor filter to each patch according to the local orientation previously computed



Please send us your code using the Teams Activity – this activity will be set soon on the Teams platform

Fingerprint classification



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Whorl



Right Loop



Left Loop

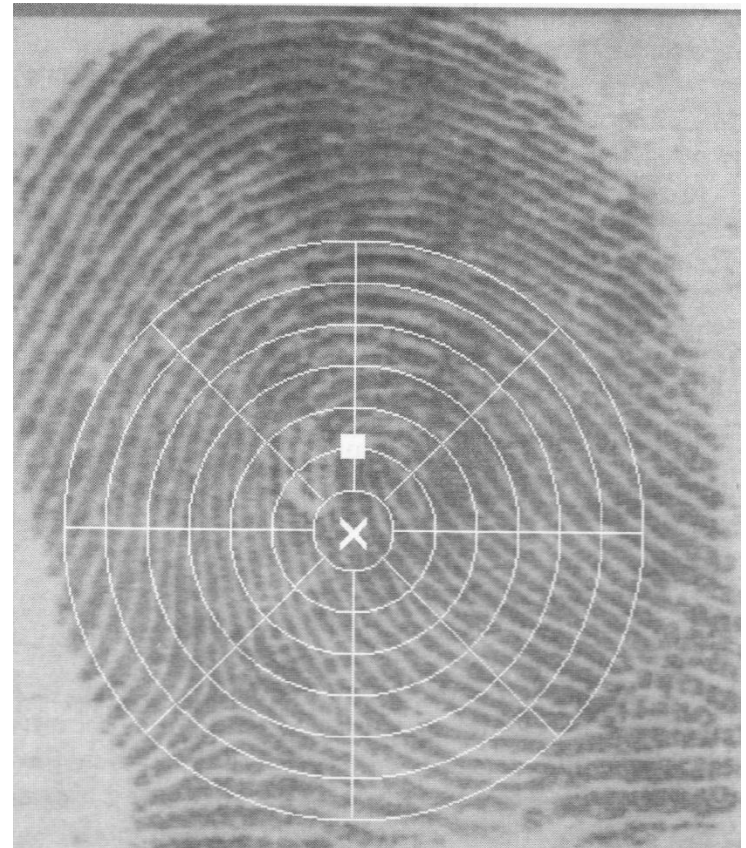


Arch

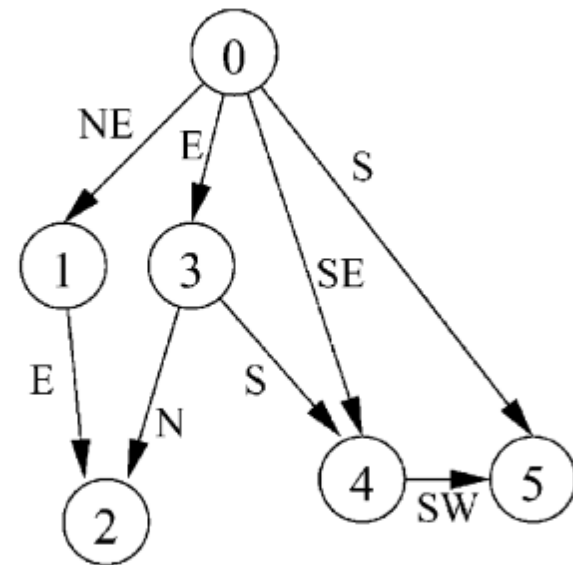
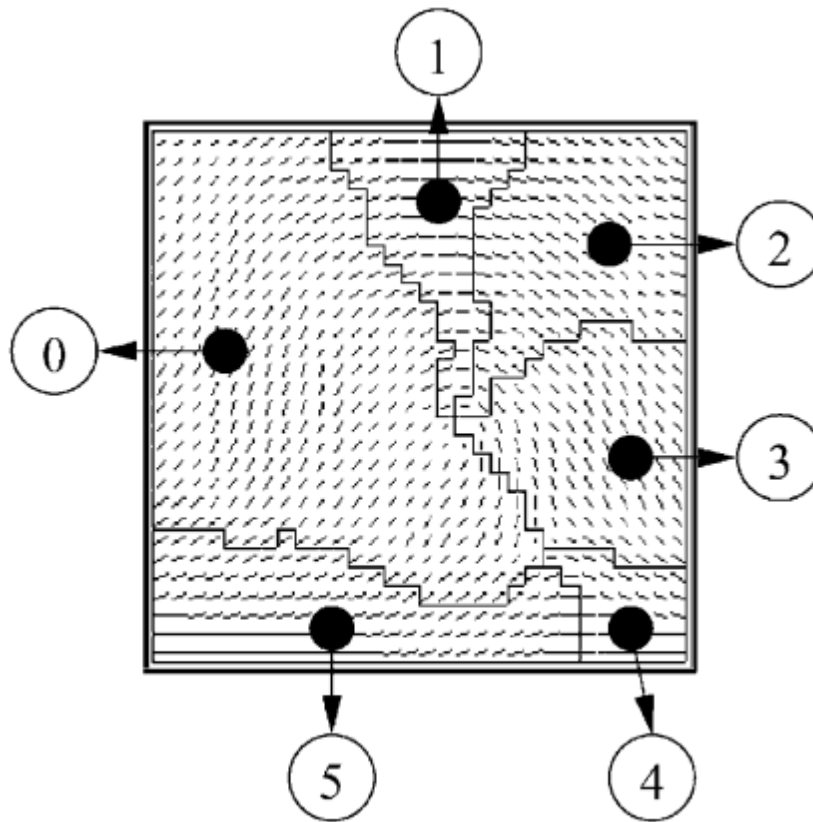


Tented Arch

- Singular points detection
 - Core and delta
- FingerCode
 - Derived from sets of Gabor filters
- Graphs
 - Relational graphs
 - DPAGs



DPAGs for fingerprint classification



K-NN

Multi-Layer Perceptron

Recursive Neural Networks (RNN)

Support Vector Machines with ECOCs

All methods estimate the posterior probability of the fingerprint class (A,L,R,T,W)

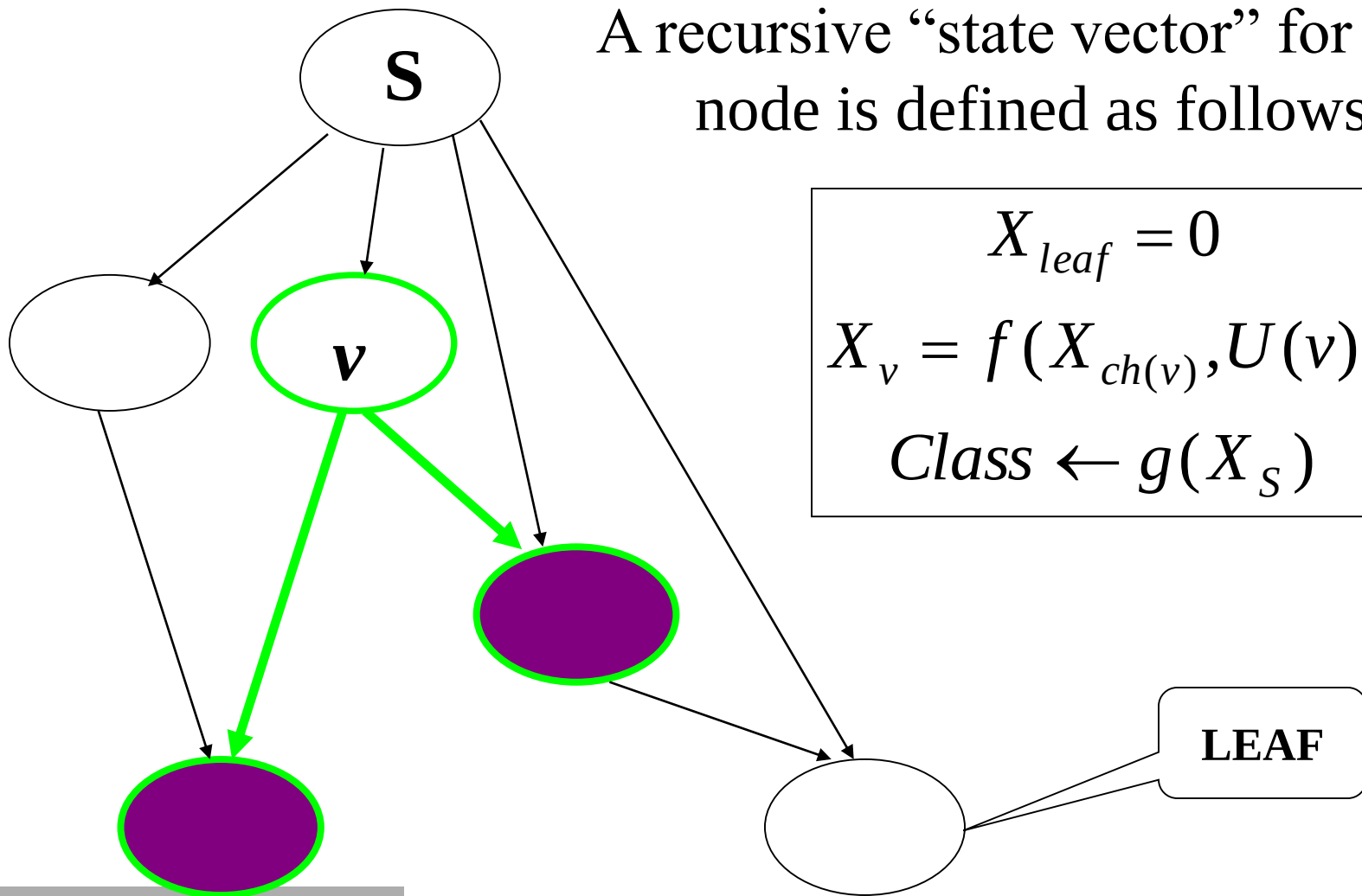
RNNs for fingerprint classification

A recursive “state vector” for each node is defined as follows:

$$X_{leaf} = 0$$

$$X_v = f(X_{ch(v)}, U(v))$$

$$Class \leftarrow g(X_S)$$

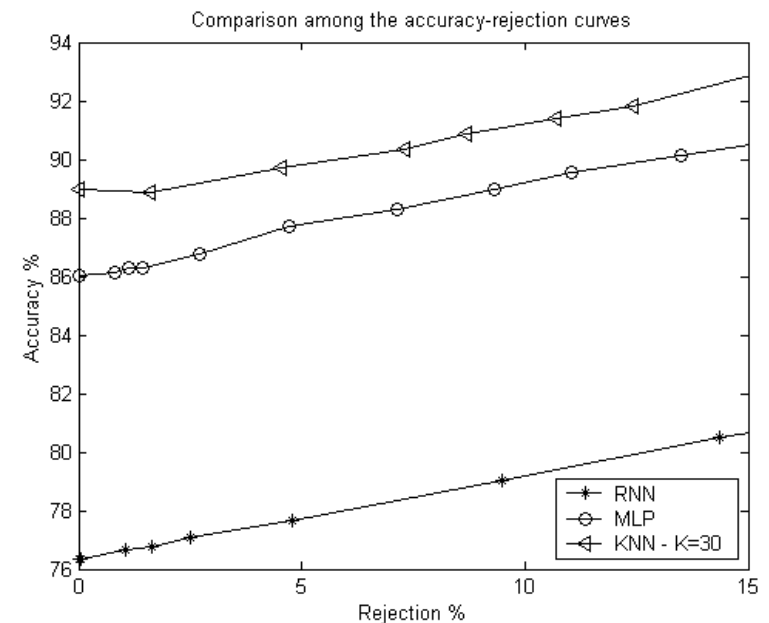


Performance evaluation

Confusion matrix

	W	R	L	A	T
W	356	23	14	3	1
R	4	344	1	7	33
L	4	2	356	6	13
A	0	2	5	371	55
T	0	7	7	48	302

Accuracy-Rejection curve



Fingerprint matching

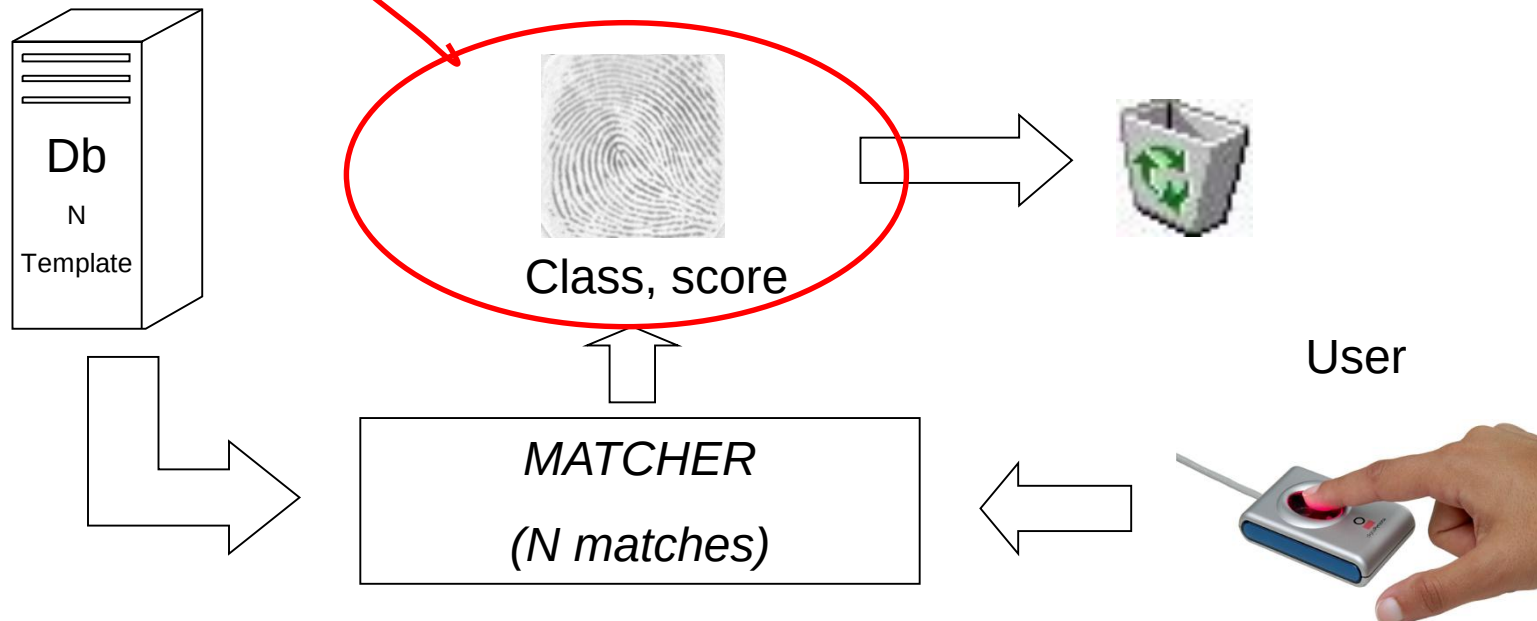


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Set of fingerprints near to the input one



ON-LINE



Correlation-based

- Spatial correlation between the input fingerprint image and the template
- Lack of robustness, computationally expensive

Filter-based

- FingerCodes

Minutiae-based

- Minutiae

Minutiae extraction



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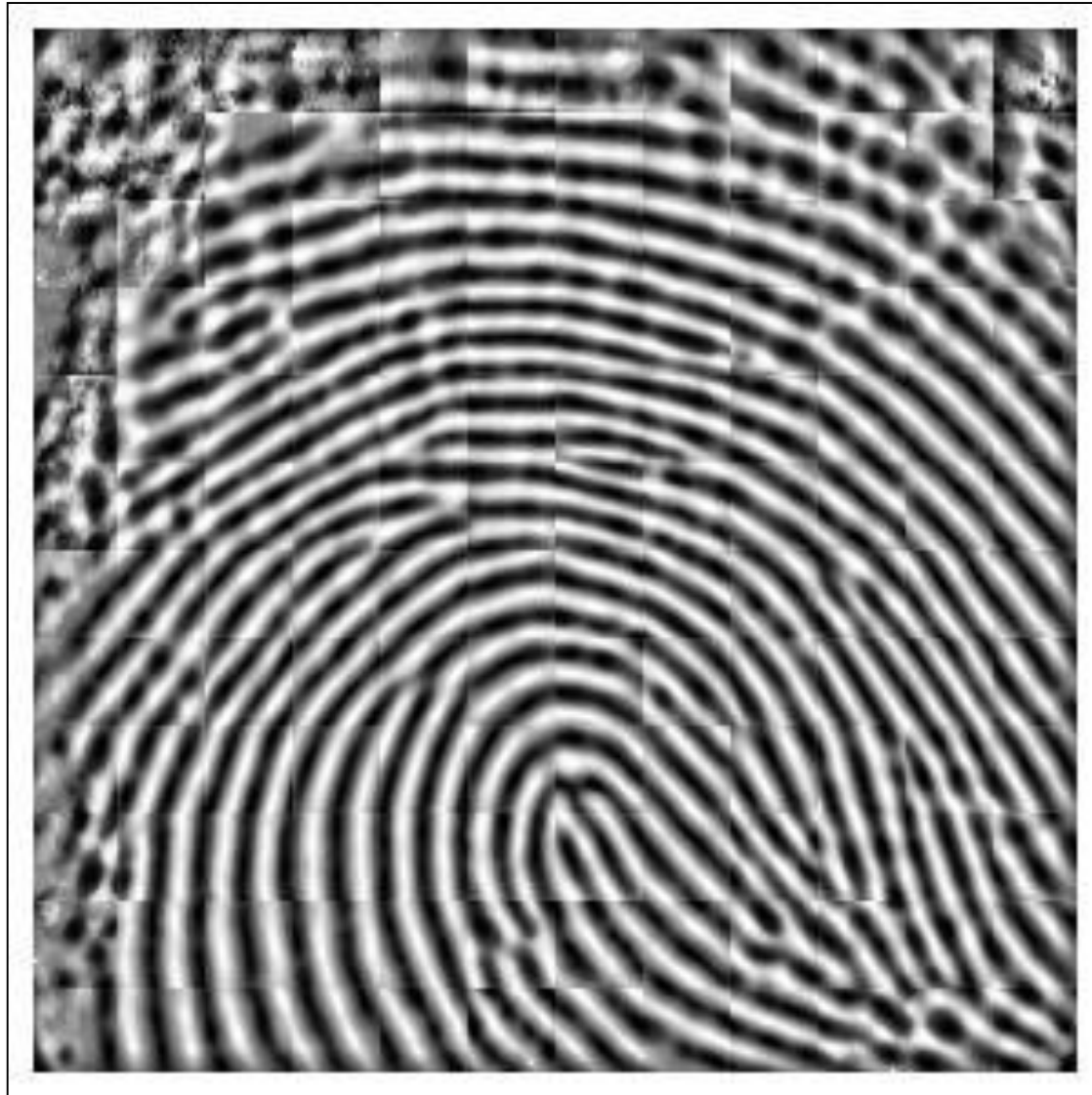
Input

Minutiae extraction



Fourier

Minutiae extraction



Rank filter

Minutiae extraction



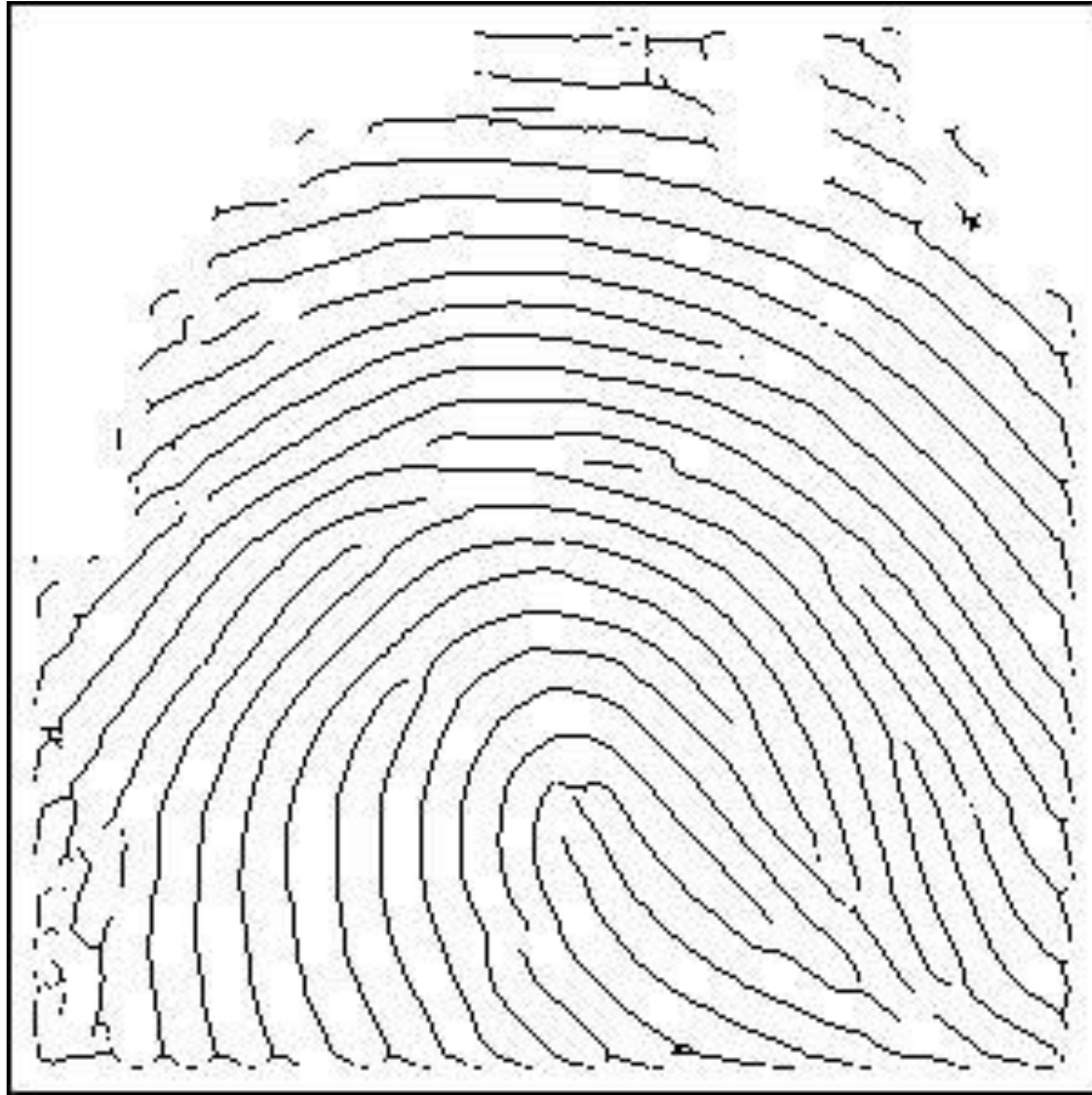
Binarization

Minutiae extraction



Thinning

Minutiae extraction



Scheletonization

Minutiae extraction



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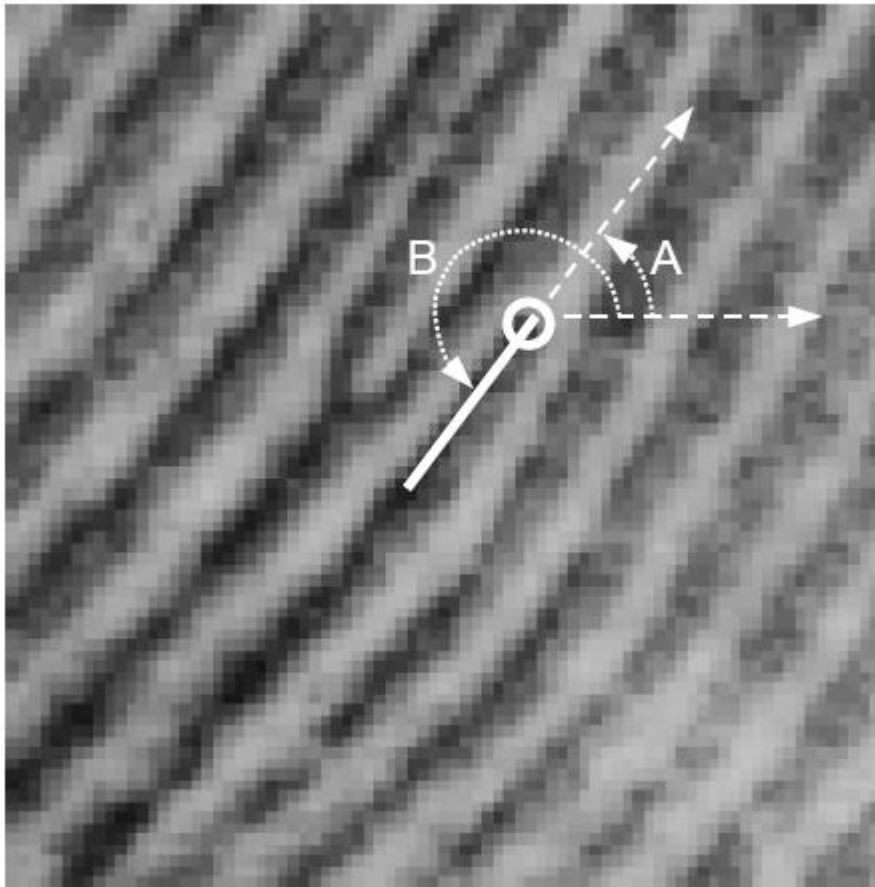


Minutiae

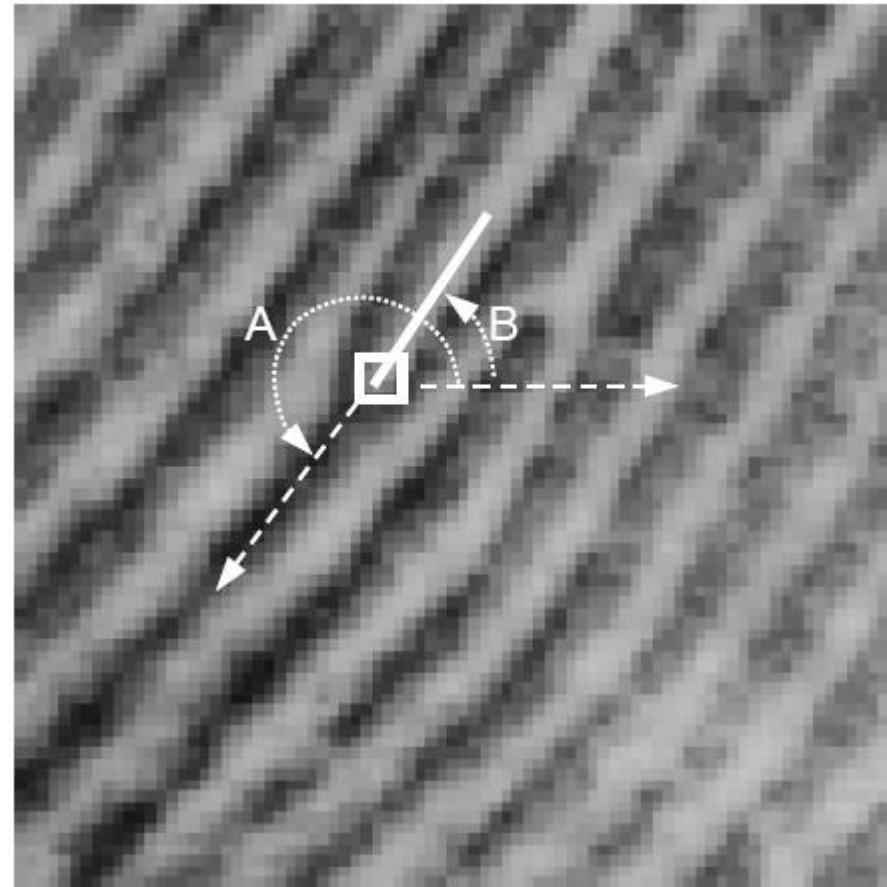
Minutiae orientation



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Ending orientation



Bifurcation orientation

Minutiae matching

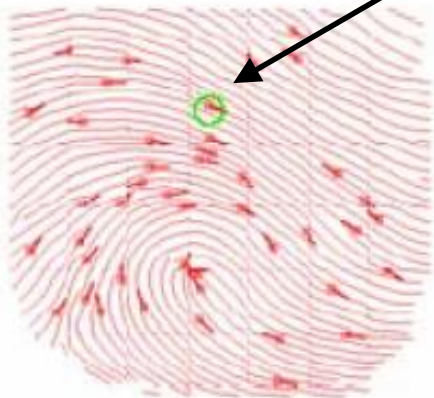


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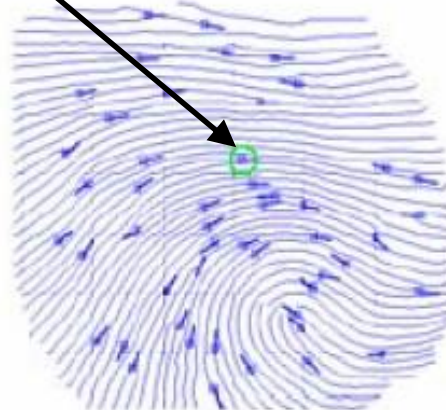
Each minutia is described by his position and orientation in the image

Since sizes of minutiae sets from two fingerprint can be different, the related representations are treated as STRINGS or RELATIONAL GRAPHS

Reference minutia

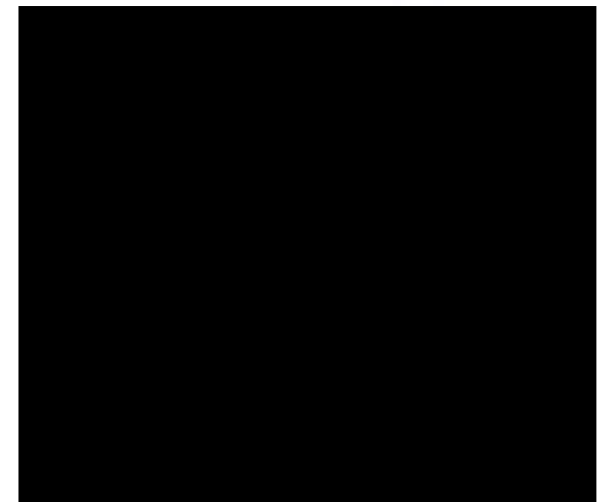


Input (n minutiae)



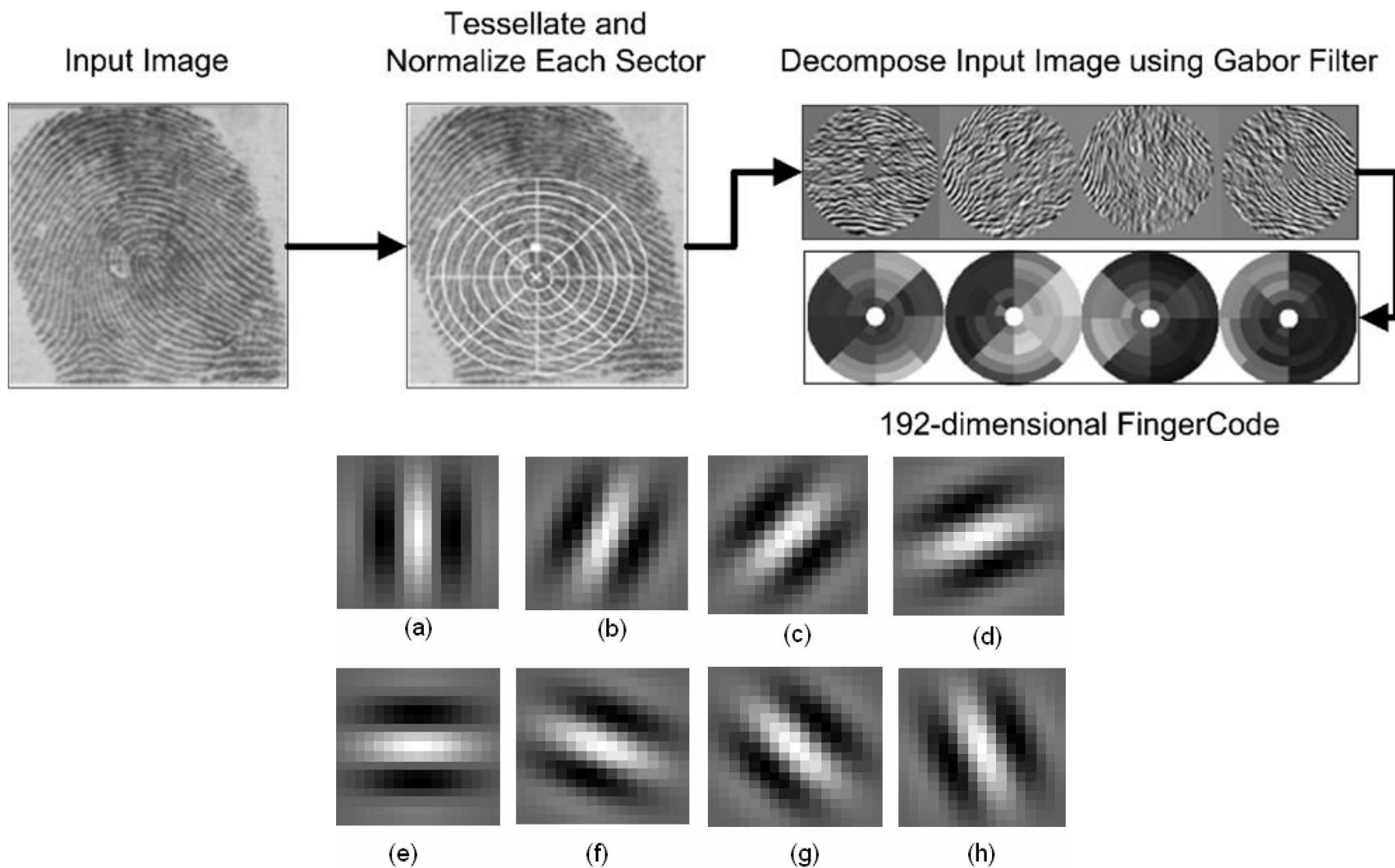
Template (m minutiae)

$$\text{score} = c^2 / (nm)$$



Match (c corresponding minutiae were found)

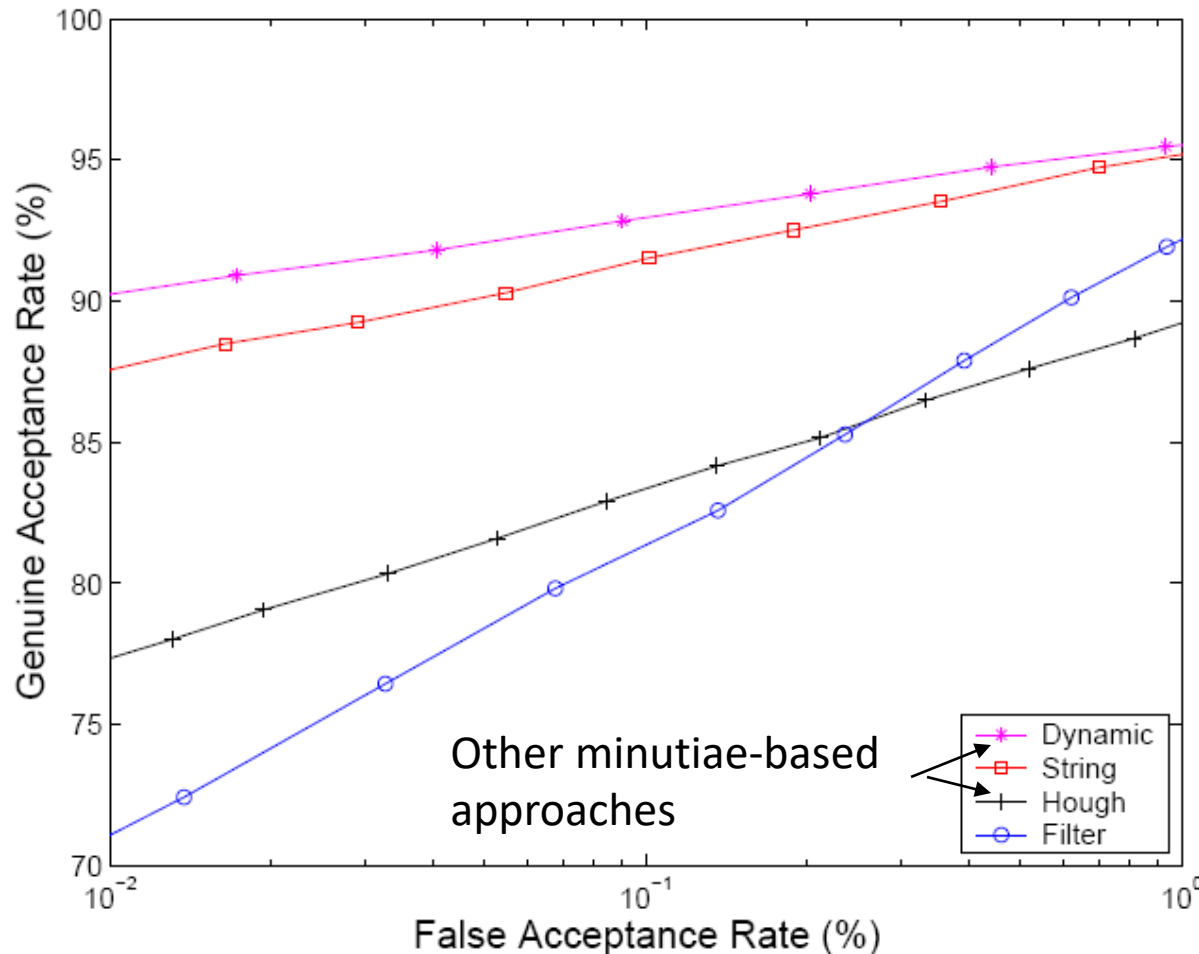
The FingerCode



ROC curves of minutiae (String) vs. fingercode (Filter) matching



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Results were obtained by adopting the Hypothesis Verification Test based on the Neyman-Pearson rule

String==Minutiae

Filter==FingerCode

Homework



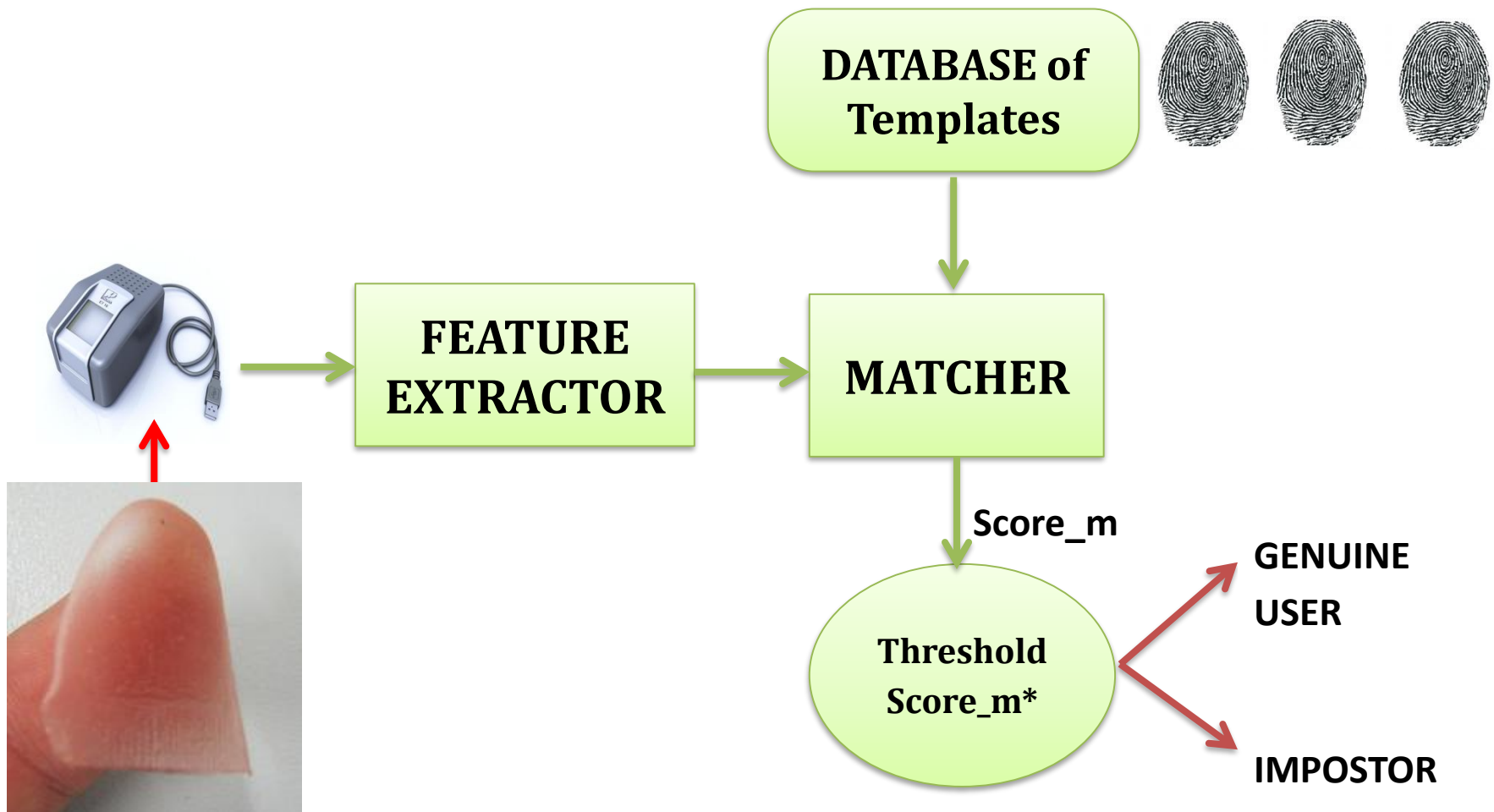
- Write your own version of a minutiae extractor (position and orientation)
- Write a possible version of minutiae matching algorithm
 - Tutorial coming soon!
- This activities will be set soon in the Teams platform with the related deadline

Fingerprint Presentation Attacks

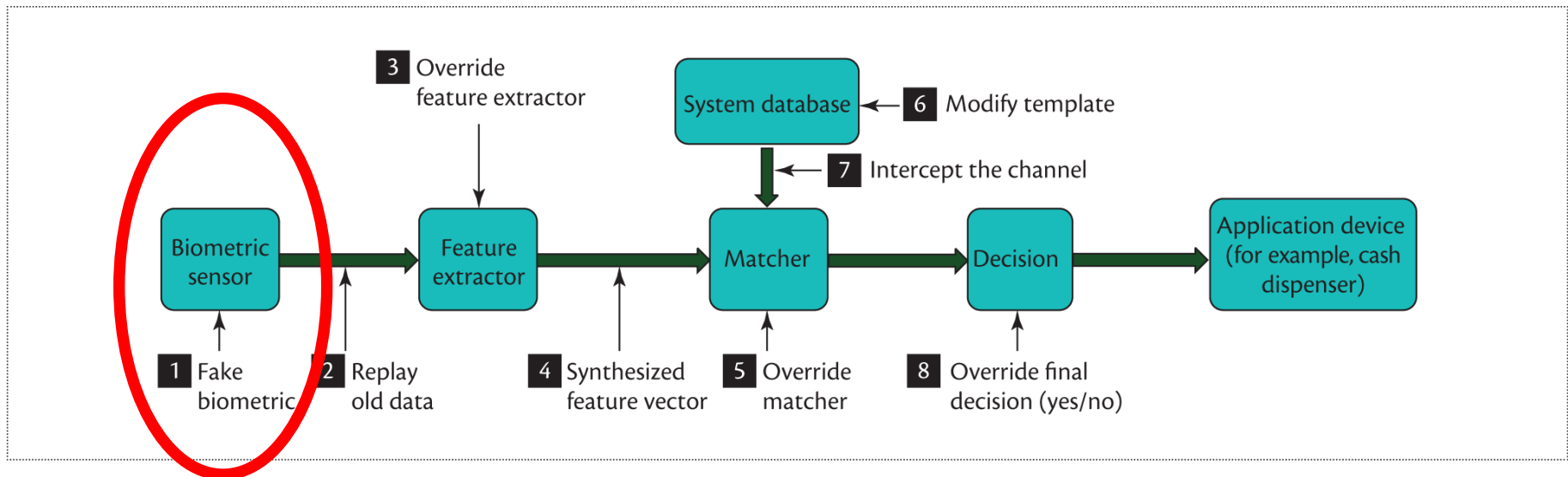


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- Also called «fake» or «spoofing» attack, it consists in submitting a replica of the biometric trait to the verification system



Vulnerability points



In the «old» days...



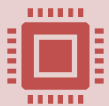
“Biometrics are hard to forge: it's hard to put a false fingerprint on your finger, or make your iris look like someone else's.



Some people can mimic others' voices, and Hollywood can make people's faces look like someone else, but these are specialized or expensive skills.



When you see someone sign his name, you generally know it is he and not someone else.”



(from “Biometrics: uses and abuses”, B. Schneier, *Inside Risk*, CACM 42, 8, 1999)

The «Prophet»: A. C. Doyle



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- S. Holmes, «The Adventure of Norwood Builder»



"Look at that with your magnifying glass, Mr. Holmes."

"Yes, I am doing so."

"You are aware that no two thumb-marks are alike?"

"I have heard something of the kind."

"That is final," said Lestrade.

"Yes, that is final," Watson involuntarily echoed.

"It is final," said Holmes.

The loss of innocence



- **The commonly adopted reference work about «fingerprint spoofing» has been published in 2002 by Matsumoto et al.**
 - Proceedings of SPIE Vol. #4677, Optical Security and Counterfeit Deterrence Techniques IV, Thursday-Friday 24-25 January 2002)
- **However, it was known that fingerprint could be replicated since from the work by van der Putte and Keuning in 2000**
 - IFIP TC8/WG8.8 *Fourth Working Conference on Smart Card Research and Advanced Applications*, pages 289-303, Kluwer Academic Publishers, 2000



(a) Live Finger

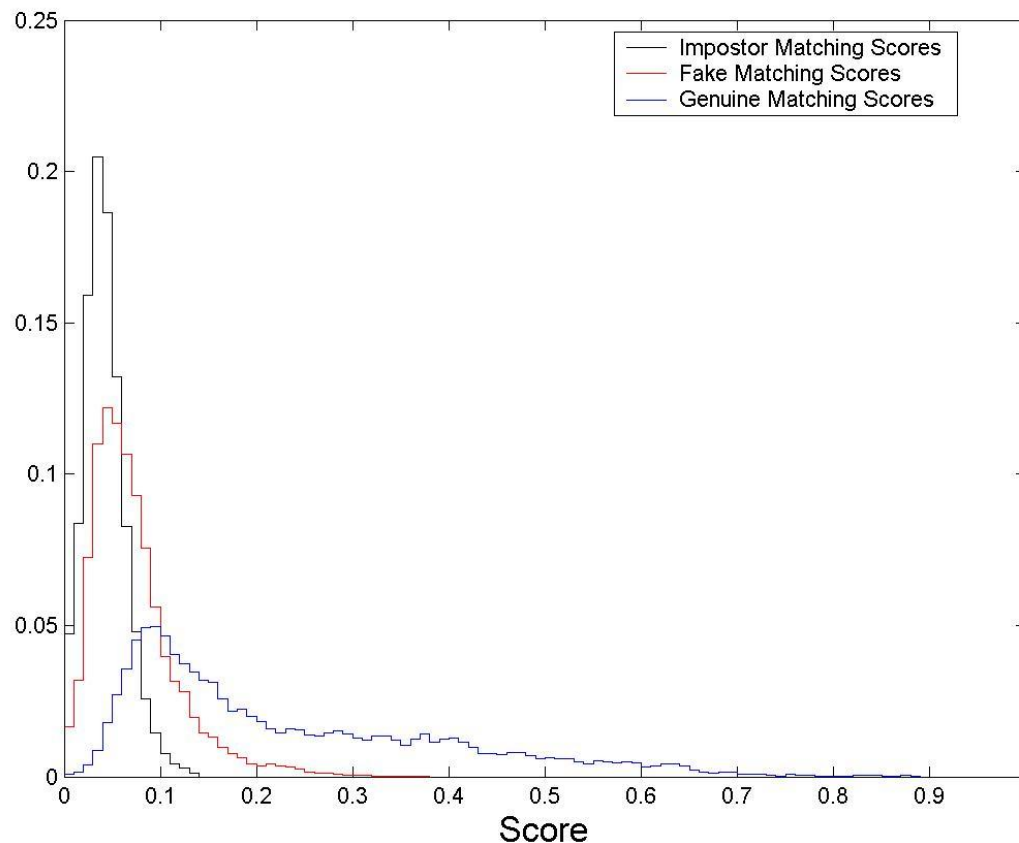
(b) Silicone Finger

(c) Gummy Finger

The loss of innocence: latent print



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Marcialis *et al.*, A Fingerprint Forensic Tool for Criminal Investigations, in C.-T. Li Ed., *Handbook of Research on Computational Forensics, Digital Crime and Investigation: Methods and Solutions*, IGI, ISBN: 978-1-60566-836-9, D.O.I. 10.4018/978-1-60566-836-9.ch002, pp. 23-52, 2010.

Work in cooperation with Ra.C.I.S. (Scientific Investigation Office of Arma dei Carabinieri, Cagliari, Italy)

Fingerprints in peril



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iPhone 5S fingerprint sensor hacked by Germany's Chaos Computer Club

Biometrics are not safe, says famous hacker team who provide video showing how they could use a fake fingerprint to bypass phone's security lockscreen

Follow Charles Arthur by email **BETA**

Charles Arthur

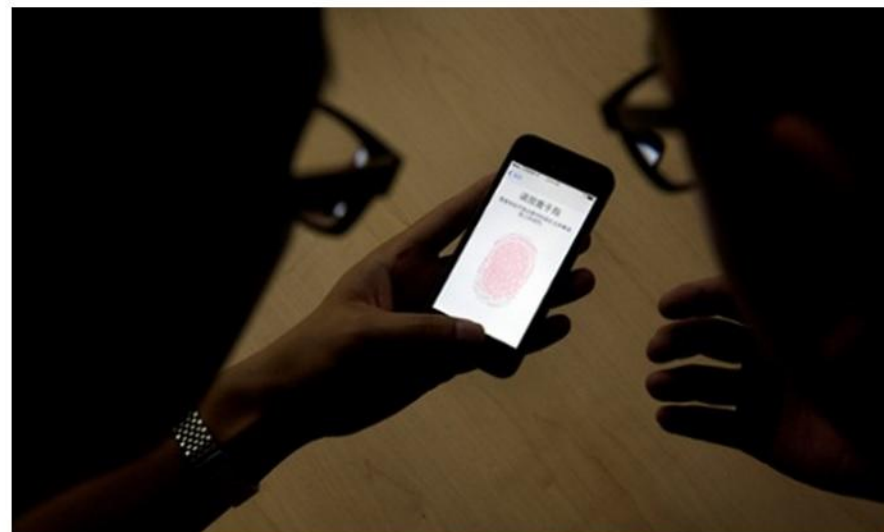
theguardian.com, Monday 23 September 2013 08.50 BST

Jump to comments (306)



Cracked Iphone by fake fingerprint

<http://www.youtube.com/watch?v=6CYtRz-H0qY>



Attacking fingerprint smartphones



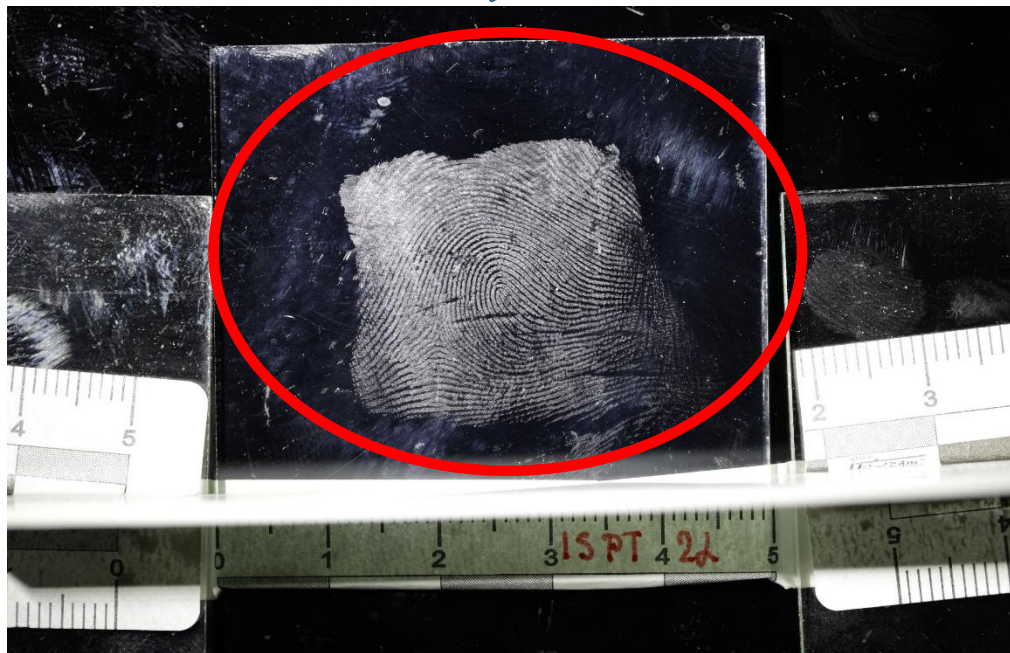
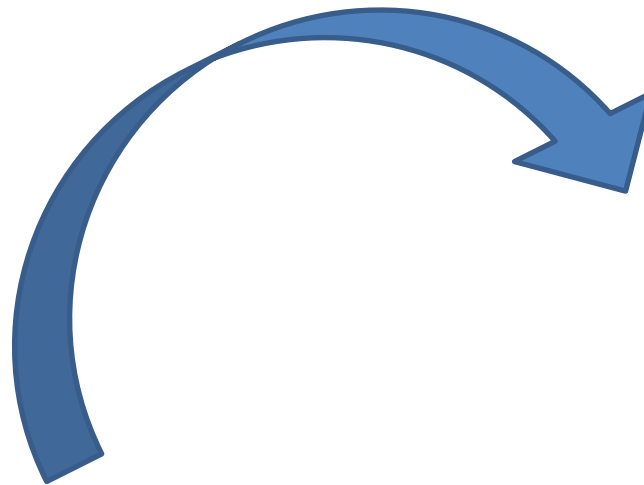
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Spoofing and latent prints



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A practical example



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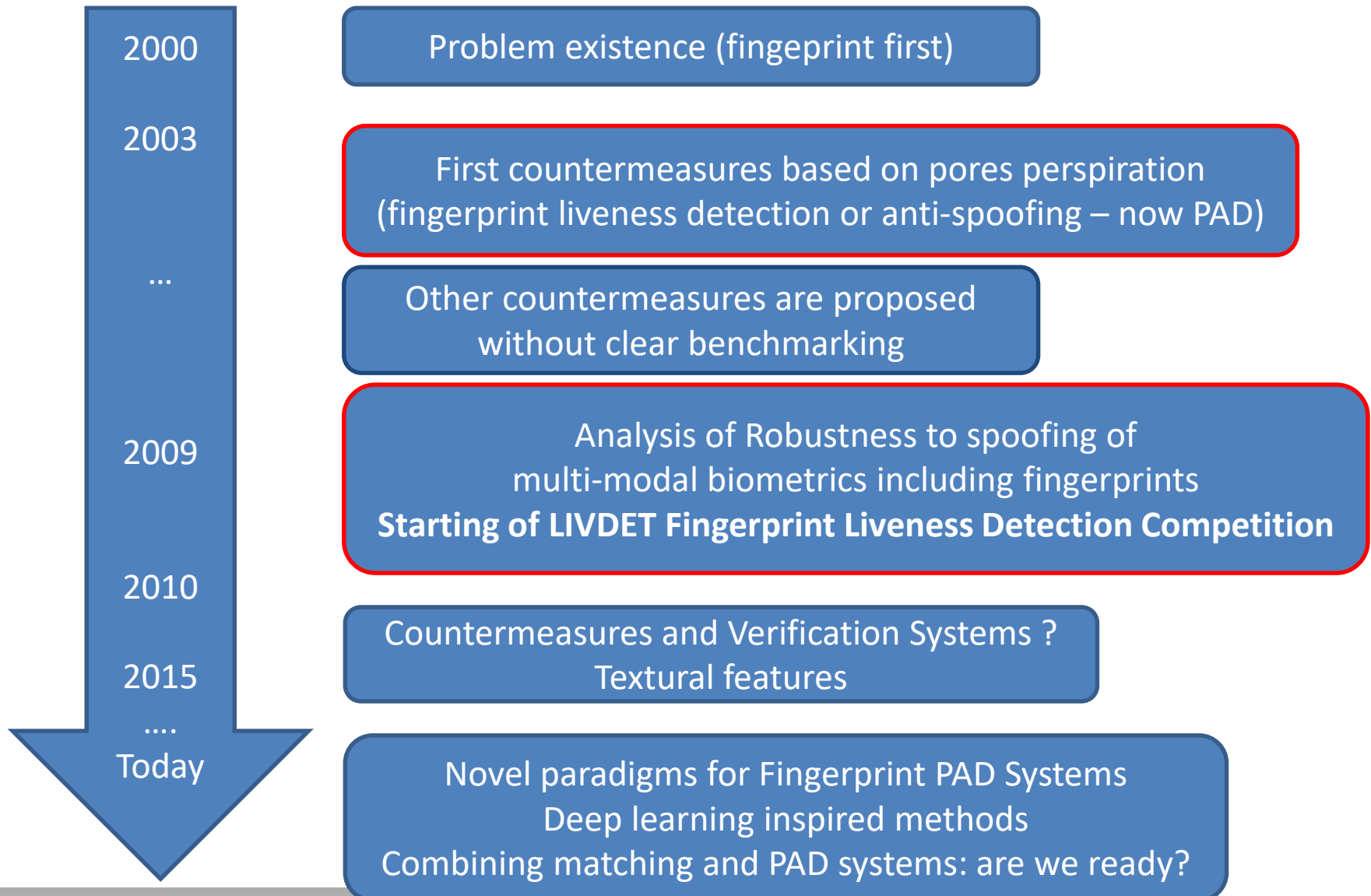


What if...

Milestones in Fingerprint PAD



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- **Fingerprint Presentation Attack Detection (also called «Liveness» or «Spoofing» detection) is intended as the ability of a hardware or software-based system of «detecting» that the fingerprint image acquired by the sensor belongs to an «alive» finger or to an artificial replica**
 - In this talk, I will only refer to software-based approaches
- **First paper on «fingerprint liveness detection» appeared in 2003 by Derakshani et al.**
 - Determination of vitality from a non-invasive biomedical measurement for use in fingerprint scanners, *Pattern Recognition*, 36 (2) (2003) 383-396
- **This concept has been generalized to any biometrics**

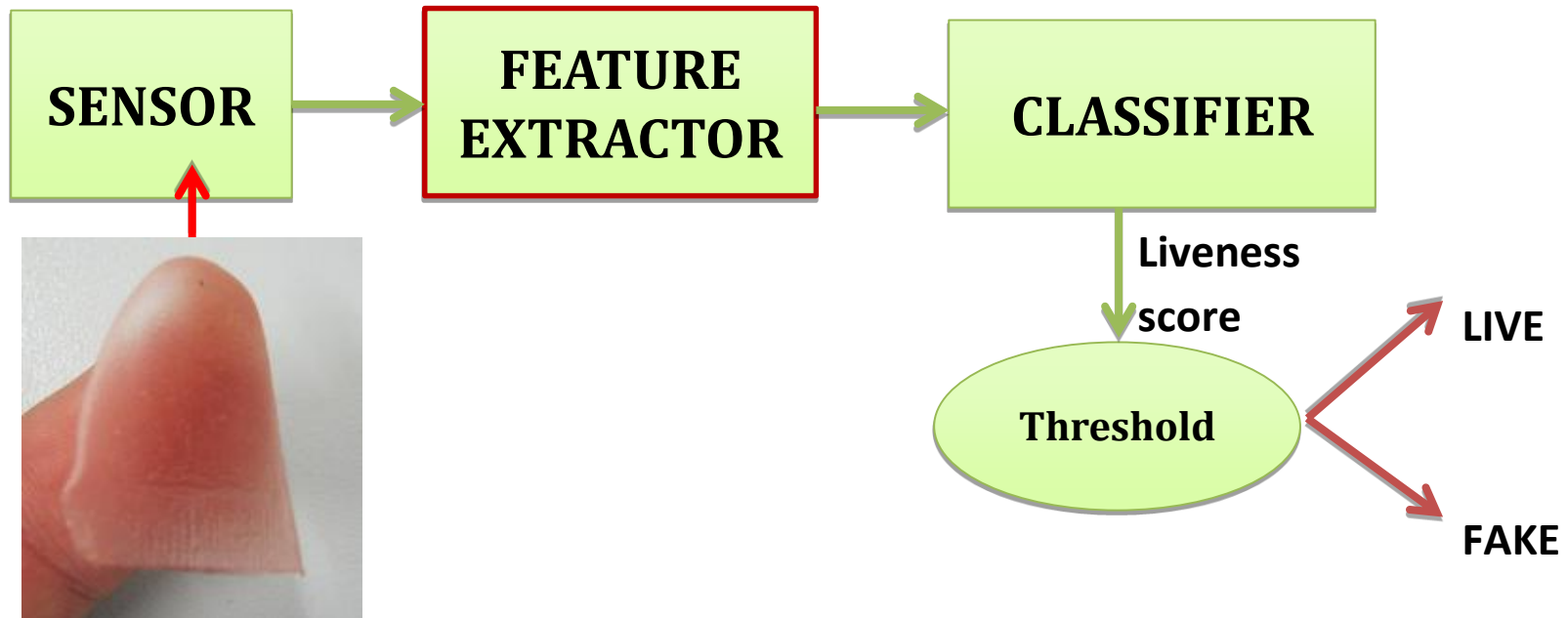
Presentation Attack Detection



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Training samples



- **ISO/IEC 30107 and ISO/IEC 19989**
- **PAI – Presentation Attack Instrument**
- **Attack presentation classification error rate (APCER)**
 - Proportion of attack presentations using the same PAI species incorrectly classified as bona fide presentations in a specific scenario
 - Commonly known as False Positive Rate (FPR), where the «positive» label is assigned to the live samples, or *bona fide* presentations
- ***Bona fide* presentation classification error rate (BPCER)**
 - Proportion of bona fide presentations incorrectly classified as attack presentations in a specific scenario
 - Commonly known as True Positive Rate (TPR)
- **Impostor attack presentation match rate (IAPMR)**
 - In a full system evaluation of a verification system, the proportion of impostor attack presentation using the same PAI species in which the target reference is matched
 - Also called Spoof-False Acceptance Rate (SFAR)

Designing a FPAD: a «tailoring» job



- **Why ?**

- It requires a specific design on the basis of:
 - The given capture device (fingerprint sensor)
 - The user population
 - The feature set adopted

- **Key points**

- Feature extraction
- Set of samples for the classifier training
 - Materials for spoofs
 - User population
 - Sensor-specific

Fake fingerprint fabrication



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Laboratorio di Impronte Digitali e Volti



Dipartimento di Ingegneria
Elettrica ed Elettronica



Pattern Recognition and
Applications Lab

Università degli Studi di Cagliari

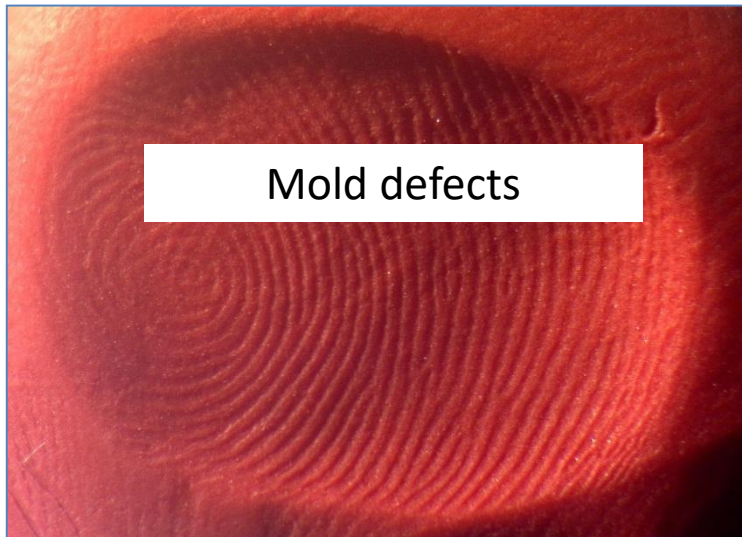
clideo.com

The features «battle»



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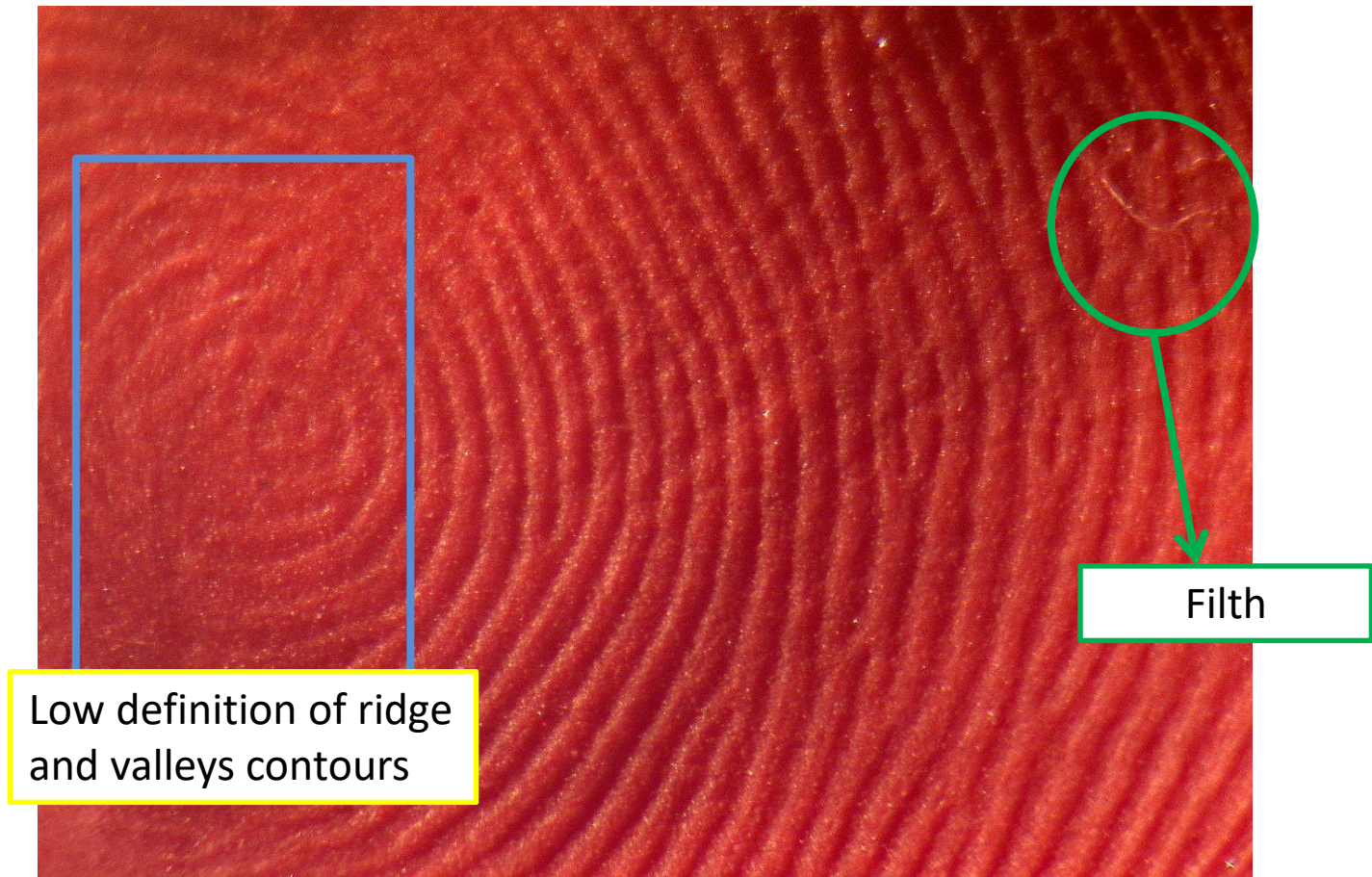
- We pointed out in our investigations that previous works were only able to look for features and measurements pointing out the «liveness» of the fingertip with respect to the «spoof», or «fake»
 - E.g. perspiration which is present, maybe, only on «live» fingers!
- But even fake fingerprint fabrication leads to some interesting differences with respect to «live» data....



Molds defects



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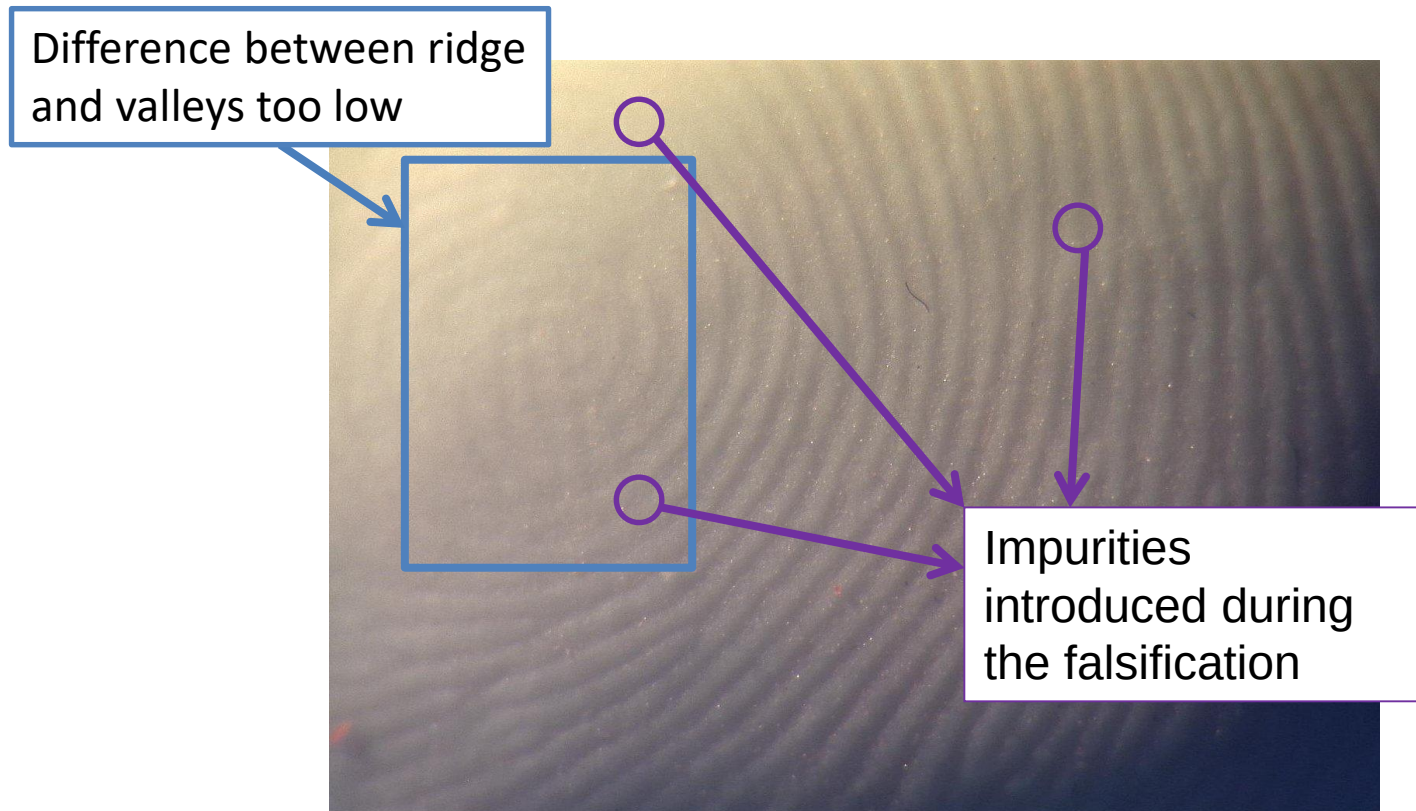


2009/2010

Cast defects



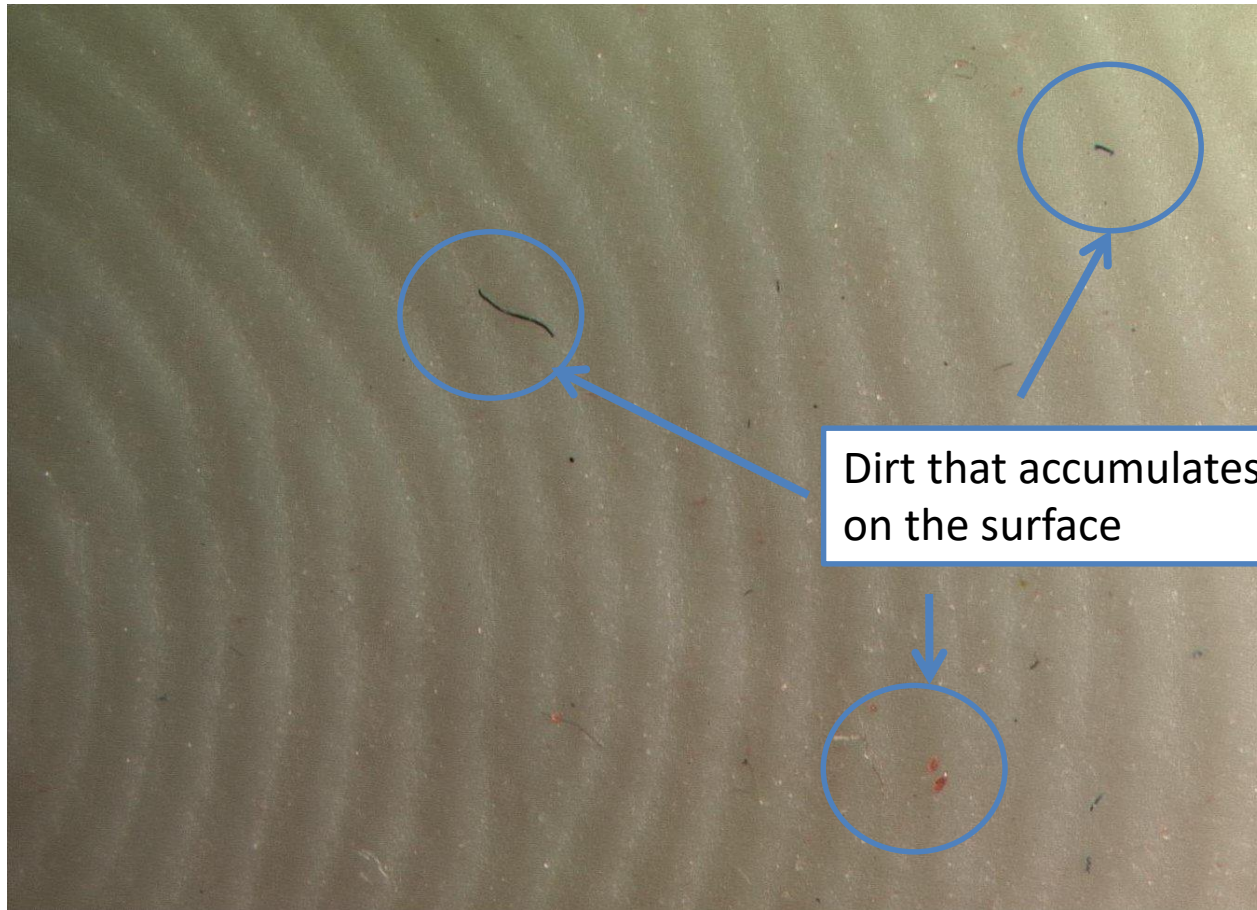
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Cast defects



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Dirt that accumulates
on the surface

Cast defects



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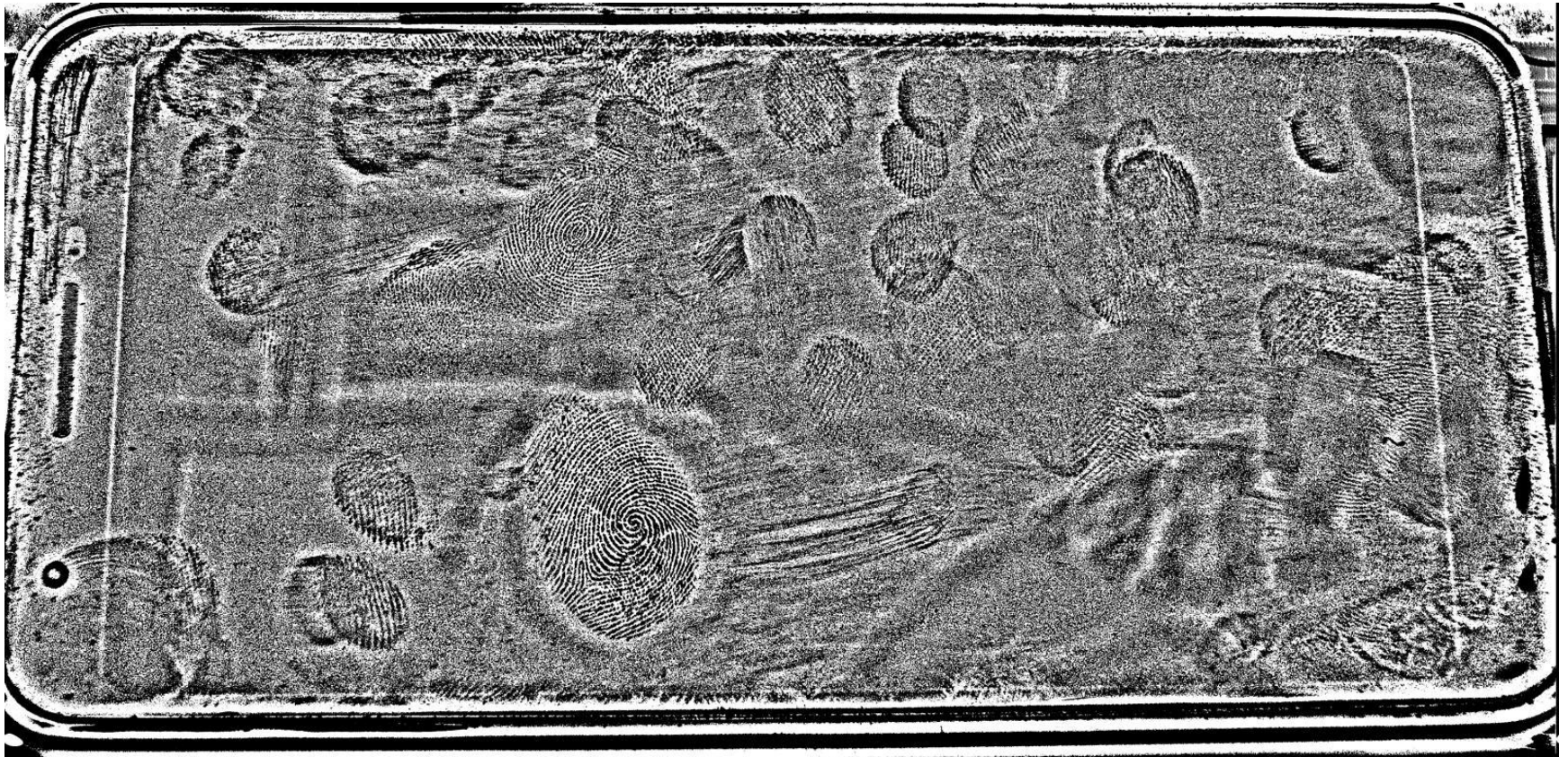


Material for the
creation of spoofs is
perishable

Things always get worse



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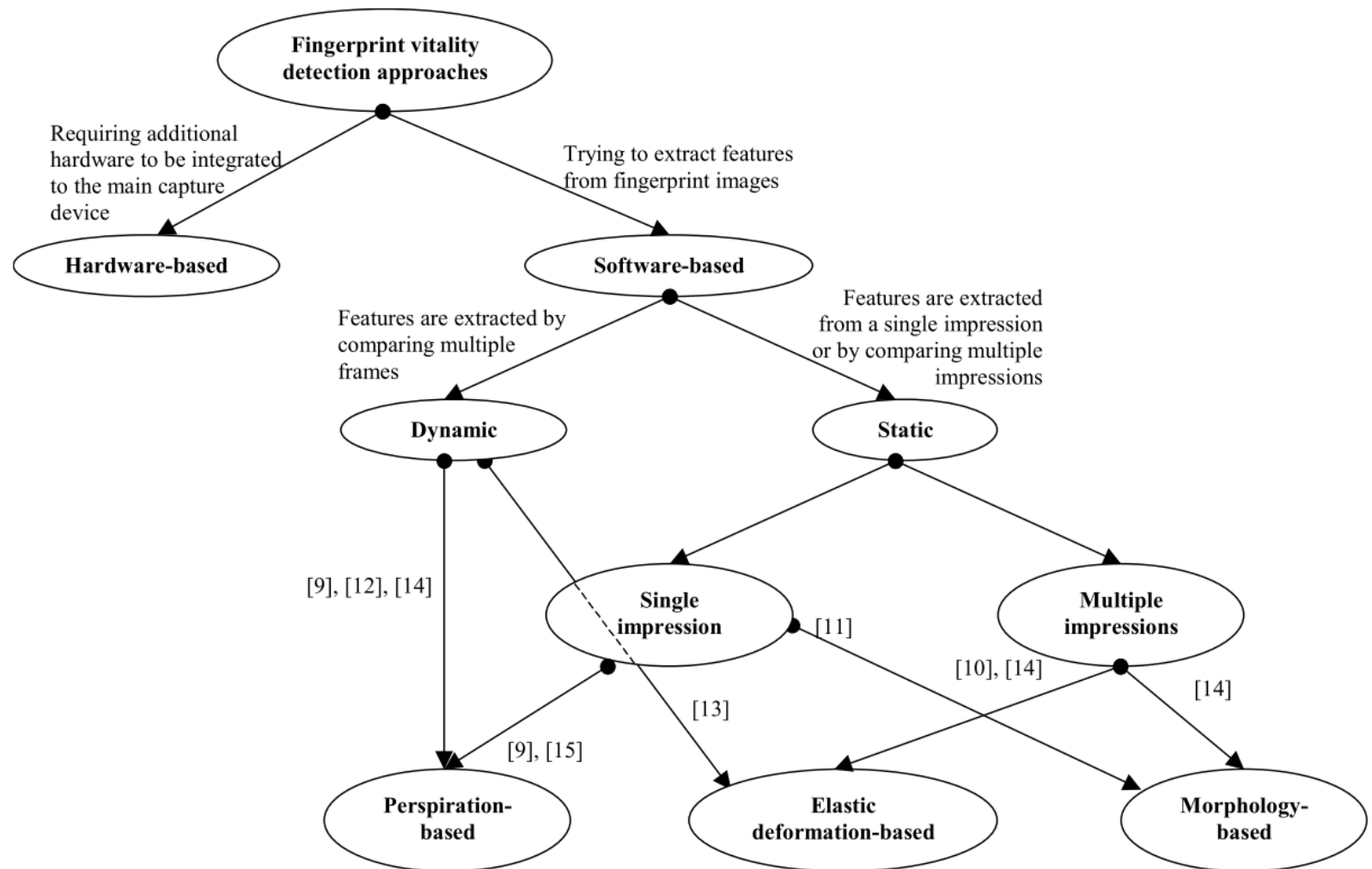


The «core»: the features extraction



- The majority of recent works on liveness detection are focused on characteristics that can't be clearly correlated with the «liveness» of the pattern but to the *artifacts* that the fabrication process leads to the final replica
- **Goal: to capture textural details by «global»...**
 - Wavelet analysis (Tan and Schuckers, CVPRW 2006),
 - Power Spectrum (Coli et al., 2007; Marcialis et al., IJDCF 2012)
 - Quality-based analysis (Galbally et al., FGCS 2012)
- **...or «local» approaches**
 - Local Binary Patterns (LBP, Nikam and Agarwal, IJB 2008)
 - Multi-Scale LBP (Jia et al., IS 2014)
 - Local Phase Quantization (LPQ, Ghiani et al., ICPR 2012)
 - Binarized Statistical Independent Features (BSIF, Ghiani et al., IET Biom. 2017)
 - Histograms of Invariant Gradient (HIG, Gottschlich et al., IJCB 2014)
 - Compared in several works (Biggio et al., IET Biom. 2014; Gottlisch, PLOS One 2016)

A simplified taxonomy



Live-based methods: perspiration detection

Live:



Spoof:



Dynamic measurements
extracted

Unreliable on large set of
images

Artificial imitation of the
perspiration due to the pores
presence

- Parthasaradhi et al., Time-Series Detection of Perspiration as a Liveness Test in Fingerprint Devices, IEEE Trans. on Systems, Man and Cybernetics, 2005
- Marcialis et al., Fingerprint Liveness Detection Based on Fake Finger Characteristics, International Journal of Digital Crime and Forensics, 2012

Live-based solutions: pores detection

Live

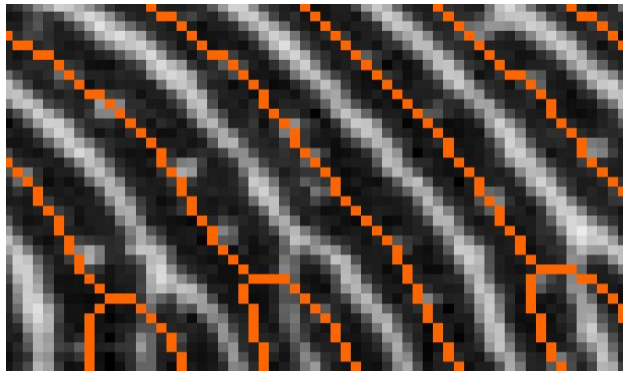


Fake



■ Extracted on:

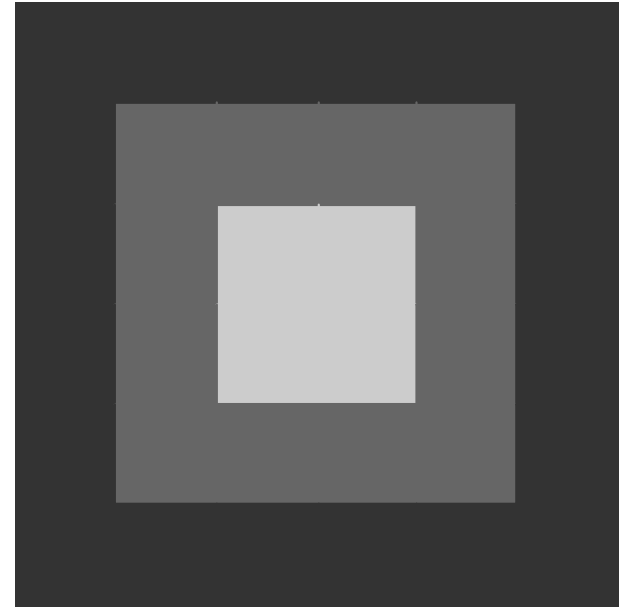
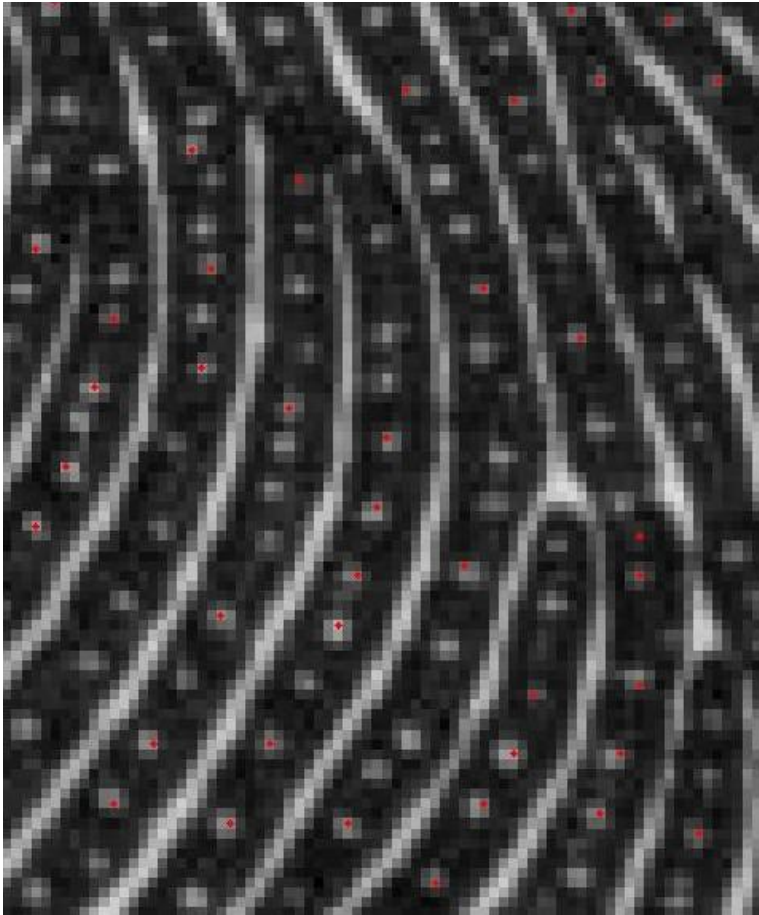
- the whole image
- a ROI 160x160 pixels around the fingerprint core
- a ROI 100x100 pixels around the fingerprint core
 - Motivation of using ROIs: pores perspiration are more evident around the core, and differences between live and fake fingerprints can be better pointed out



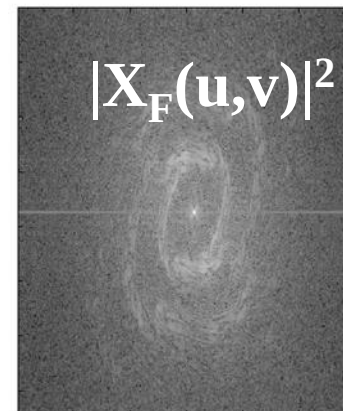
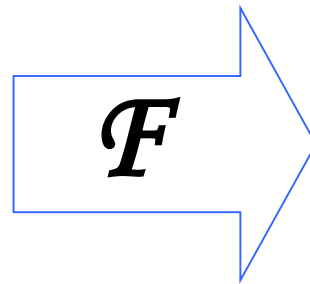
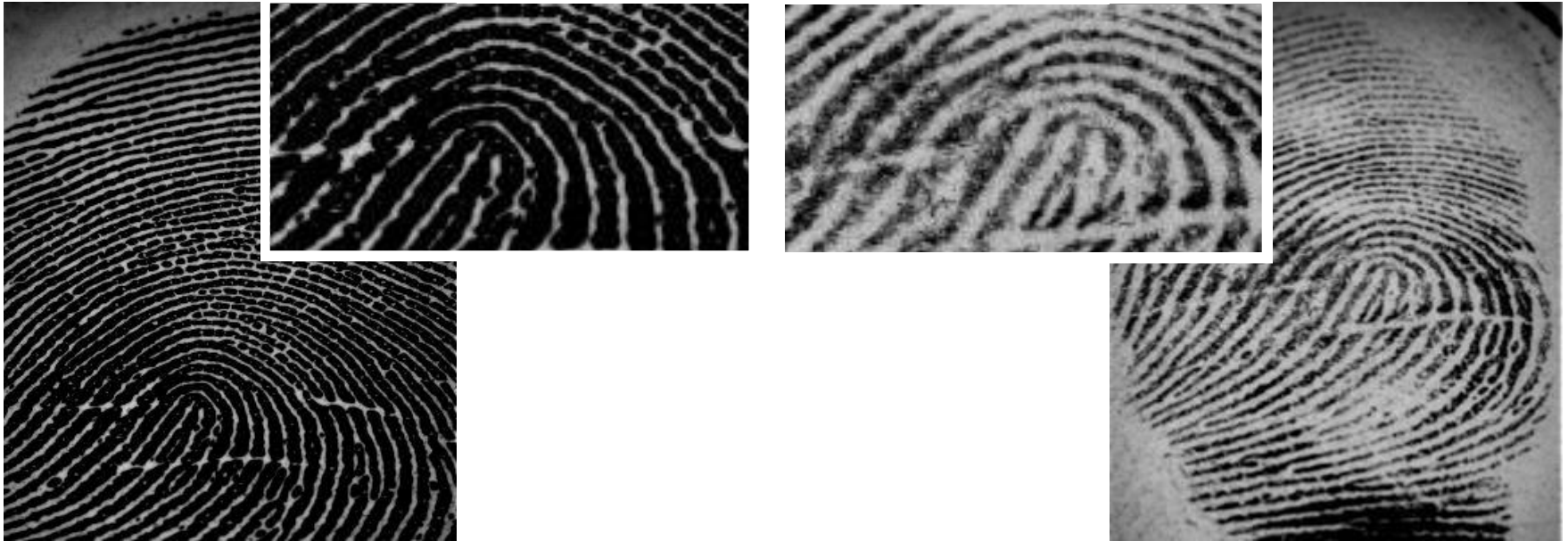
- Quality (NIST nfiq algorithm)
- Frequency of pores
- Average distance between pores
- Difference of pores number between frames (1st-5th sec)

- Marcialis et al., Analysis of Fingerprint Pores for Vitality Detection, ICPR 2010
- Johnson and Schuckers, Fingerprint Pore Characteristic for Fingerprint Liveness Detection, BioSig 2014

Simplified physiognomy of the pores



Fake-based methods: power spectrum





- The most of them describe each pixel neighboring by a binary code obtained by convolution of the image with a *manually* predefined set of filters. For example:
 - LBP works in the image domain
 - LPQ works in the frequency domain
- Binarized Statistical Image Features* (BSIF) generalize this concept in the image domain, by applying a «learning» step to derive a *statistical meaningful* set of filters

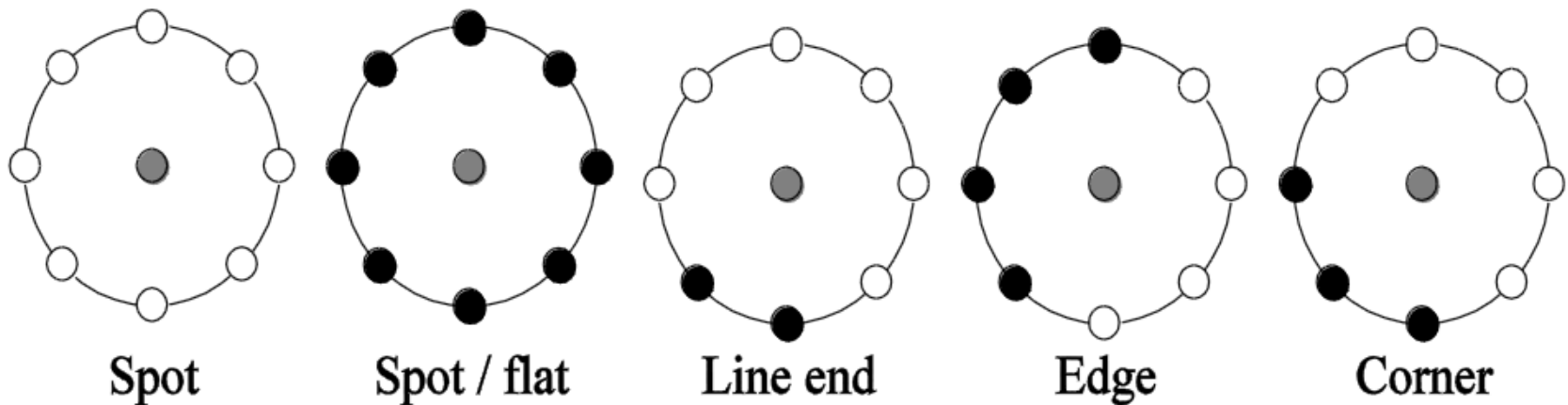
*Kannla and Rahtu, ICPR 2012

Basic idea: Local Binary Patterns



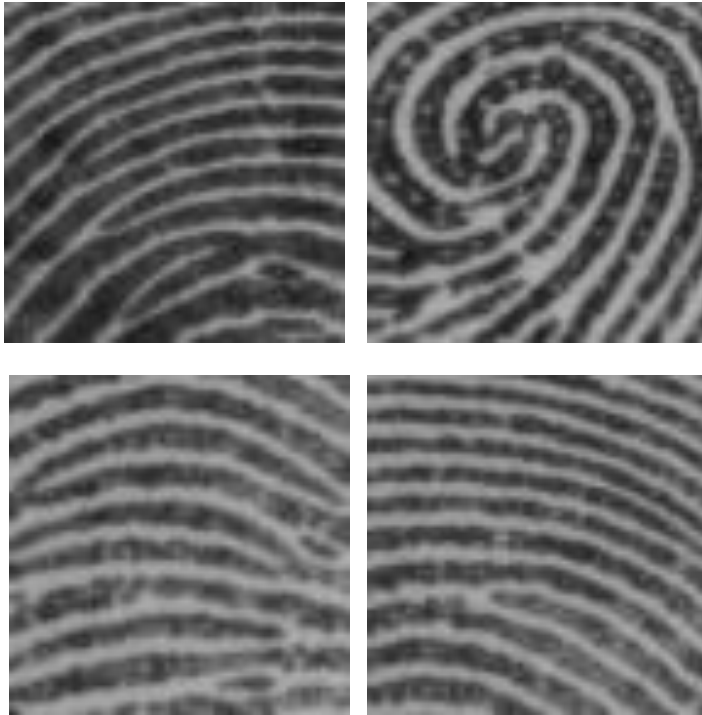
- 256 basic configurations of a 8 neighborhood along each pixels are considered and their frequencies computed
- Some of the above configurations can be referred to a «textural templates», the others can be referred to noise, thus reducing the feature vector length

1 = black
0 = white

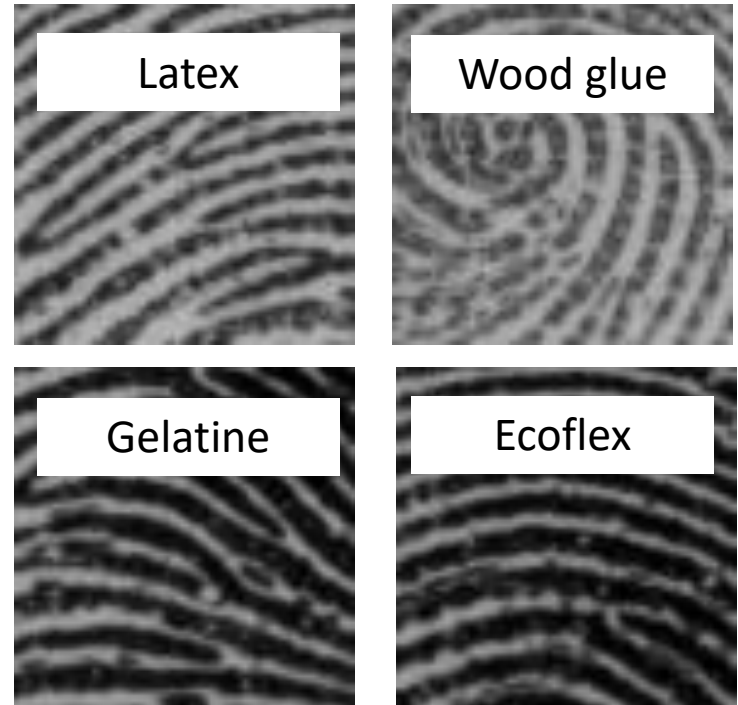


Generalization: learning filters from images

Live

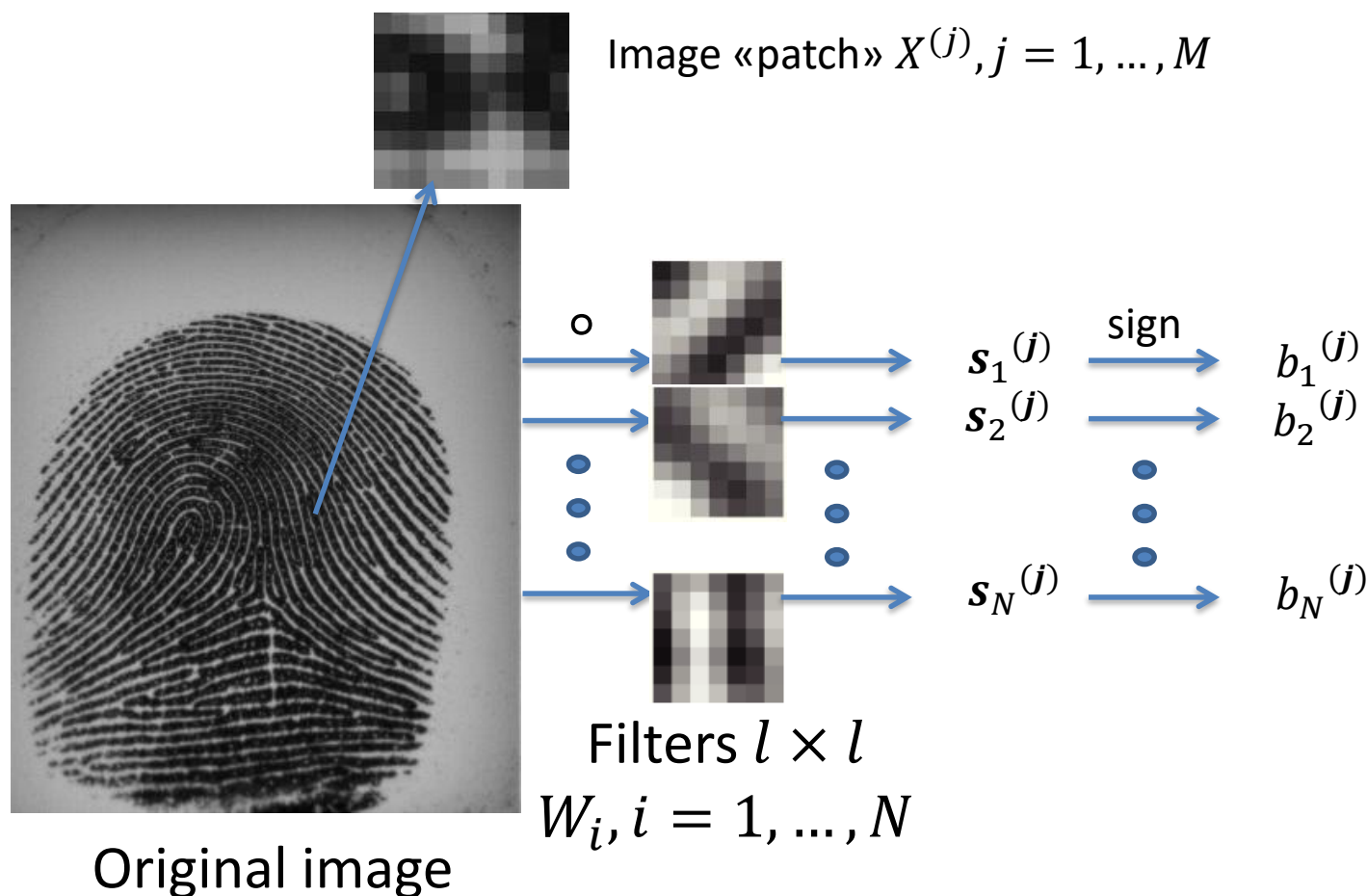


Fake



- *Grey level distributions* of live and fake images exhibit strong *local* variations that cannot be pointed out by visual inspection
- Differences are very difficult to capture by setting a *pre-defined filter set* once for all (as for LBP, LPQ)

BSIF algorithm: rationale

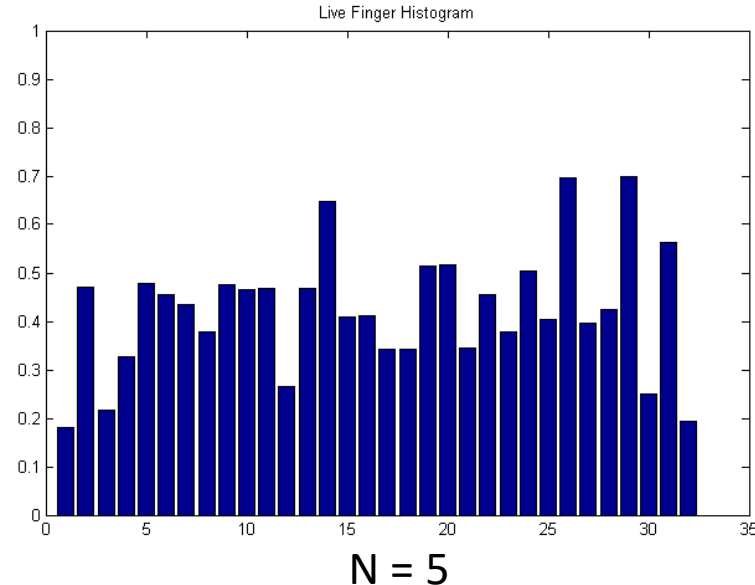


Size (l) and number (N) of filters are two «free» parameters of the algorithm
 M is the image size.

BSIF algorithm



$$\begin{aligned} B^{(1)} &= \{b_1^{(1)}, \dots, b_N^{(1)}\} \longrightarrow \\ B^{(2)} &= \{b_1^{(2)}, \dots, b_N^{(2)}\} \longrightarrow \\ &\vdots \\ B^{(M)} &= \{b_1^{(M)}, \dots, b_N^{(M)}\} \longrightarrow \end{aligned}$$



- A normalized histogram of the occurrence of all $B^{(j)}$ is obtained and used as feature vector
- The process is similar to that of LPQ computation in the frequency domain, where the local Fourier transform is sampled according to four frequency values

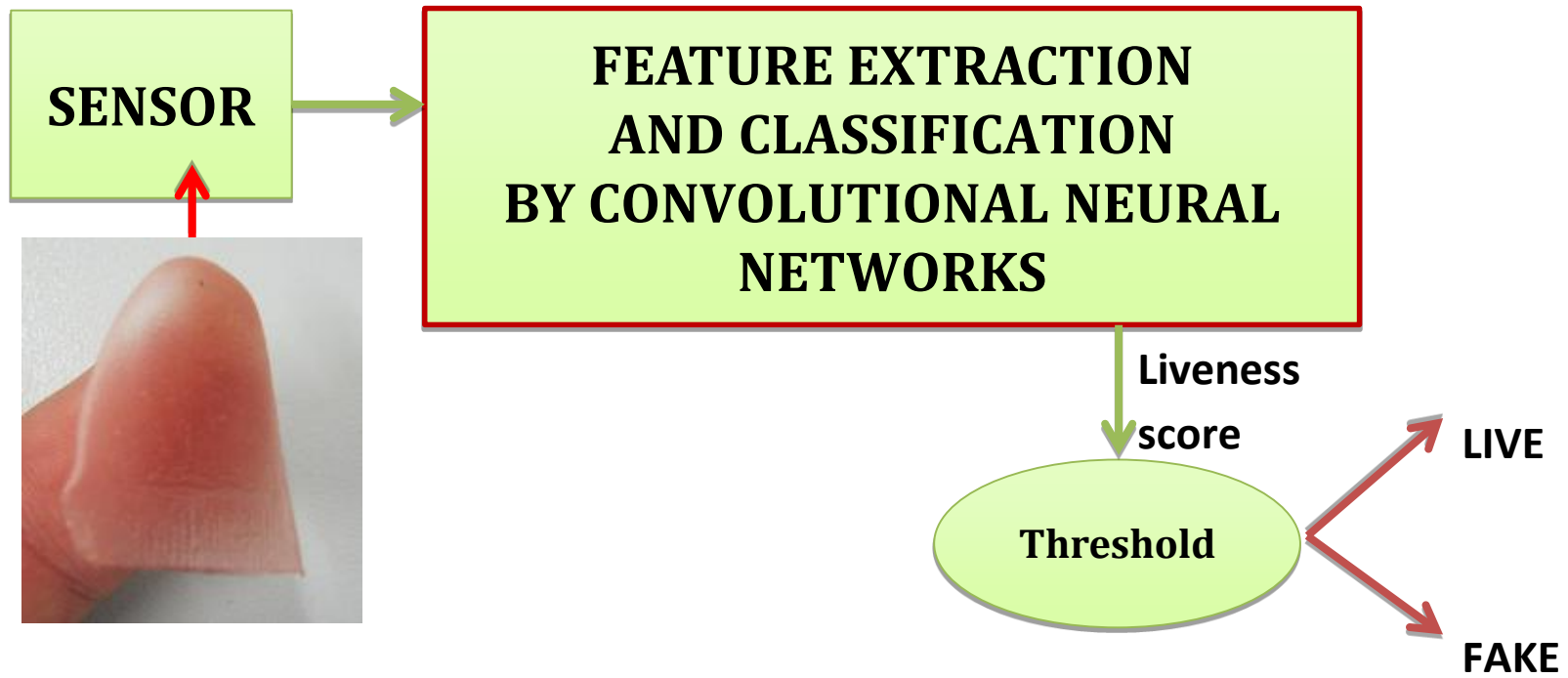
Deep networks for fingerprint PAD



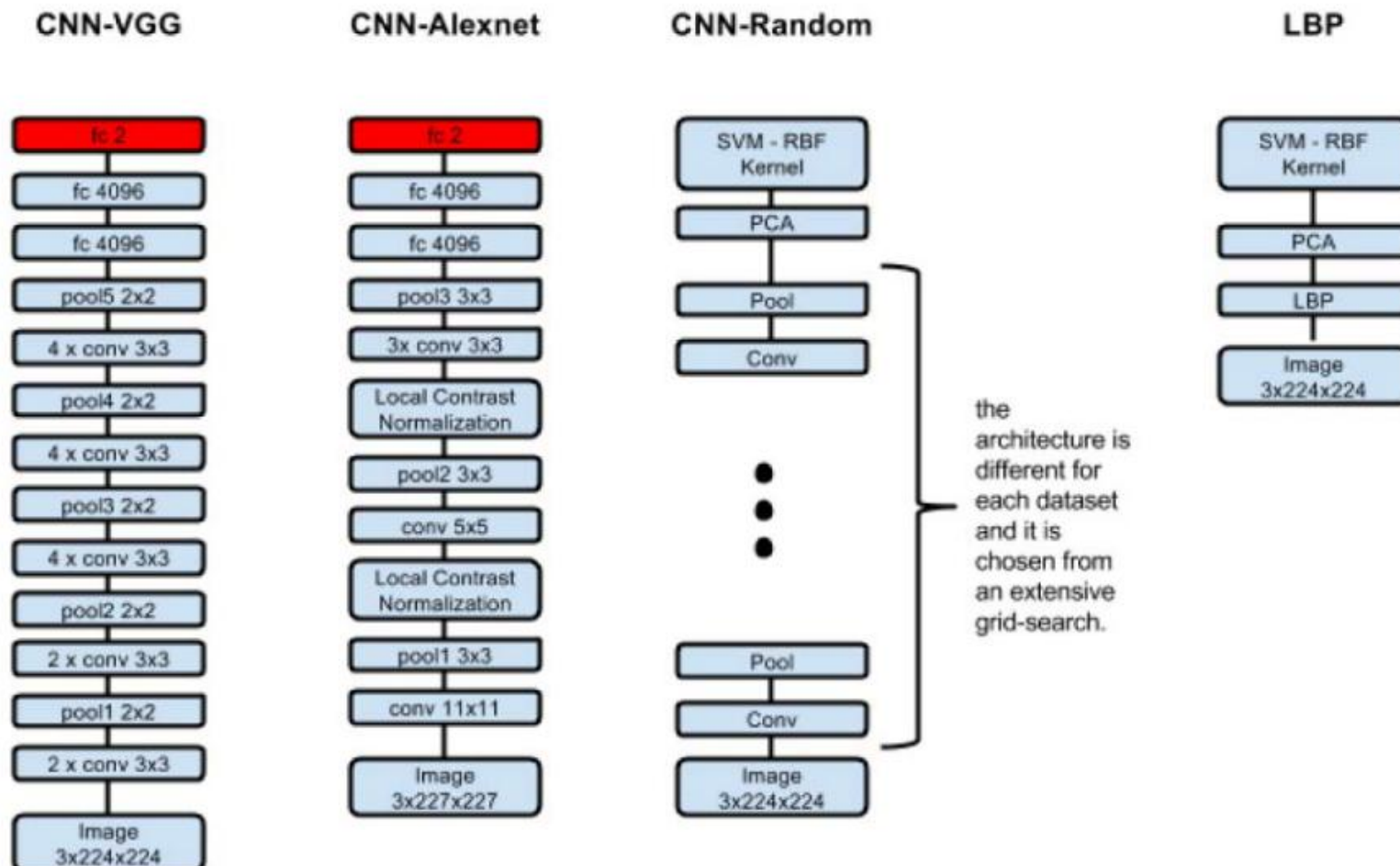
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Training samples



Just some examples

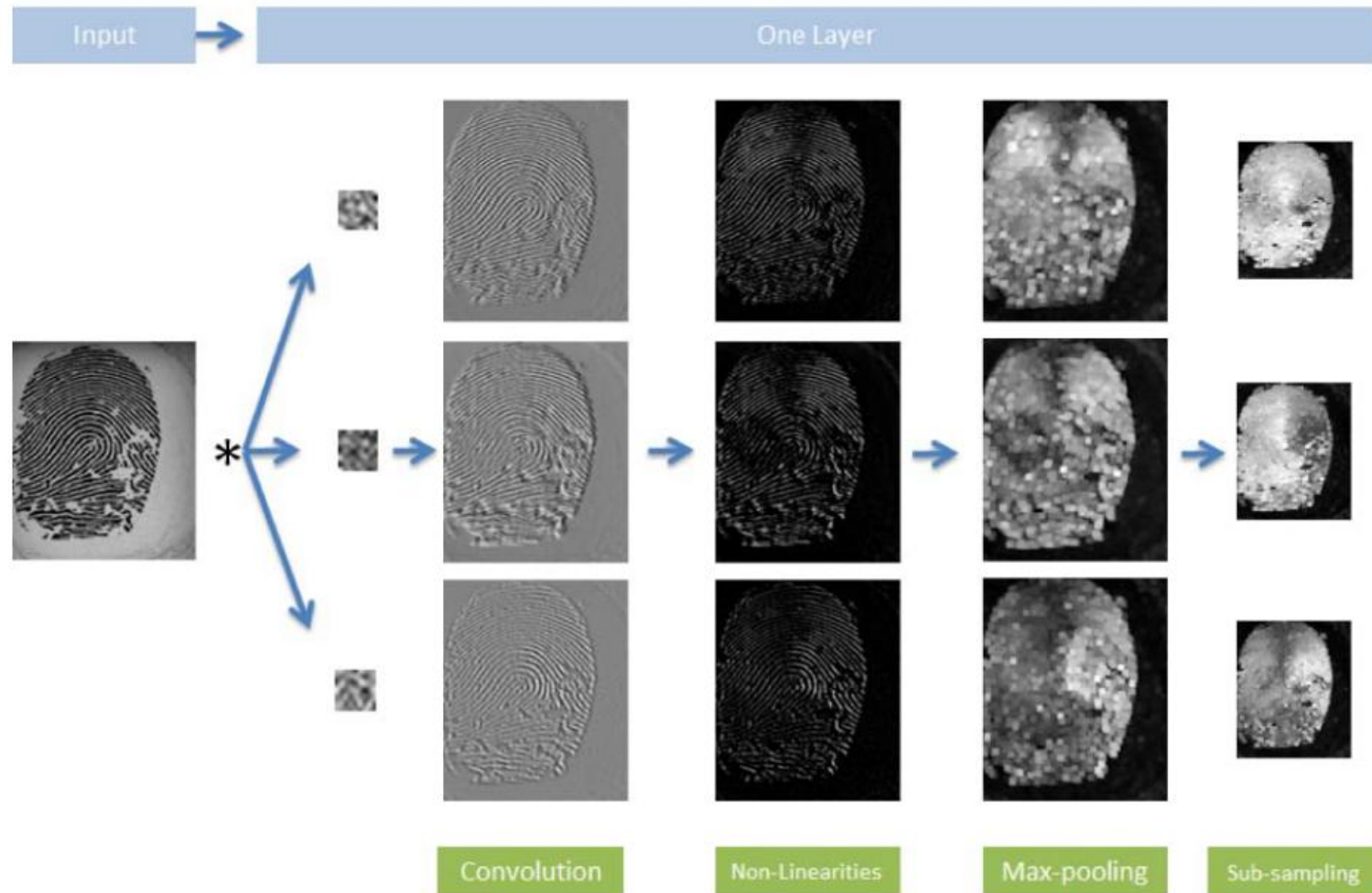


Nogueira et al., Fingerprint Liveness Detection Using Convolutional Neural Networks, IEEE TIFS 2016

CNN-based Fingerprint PAD



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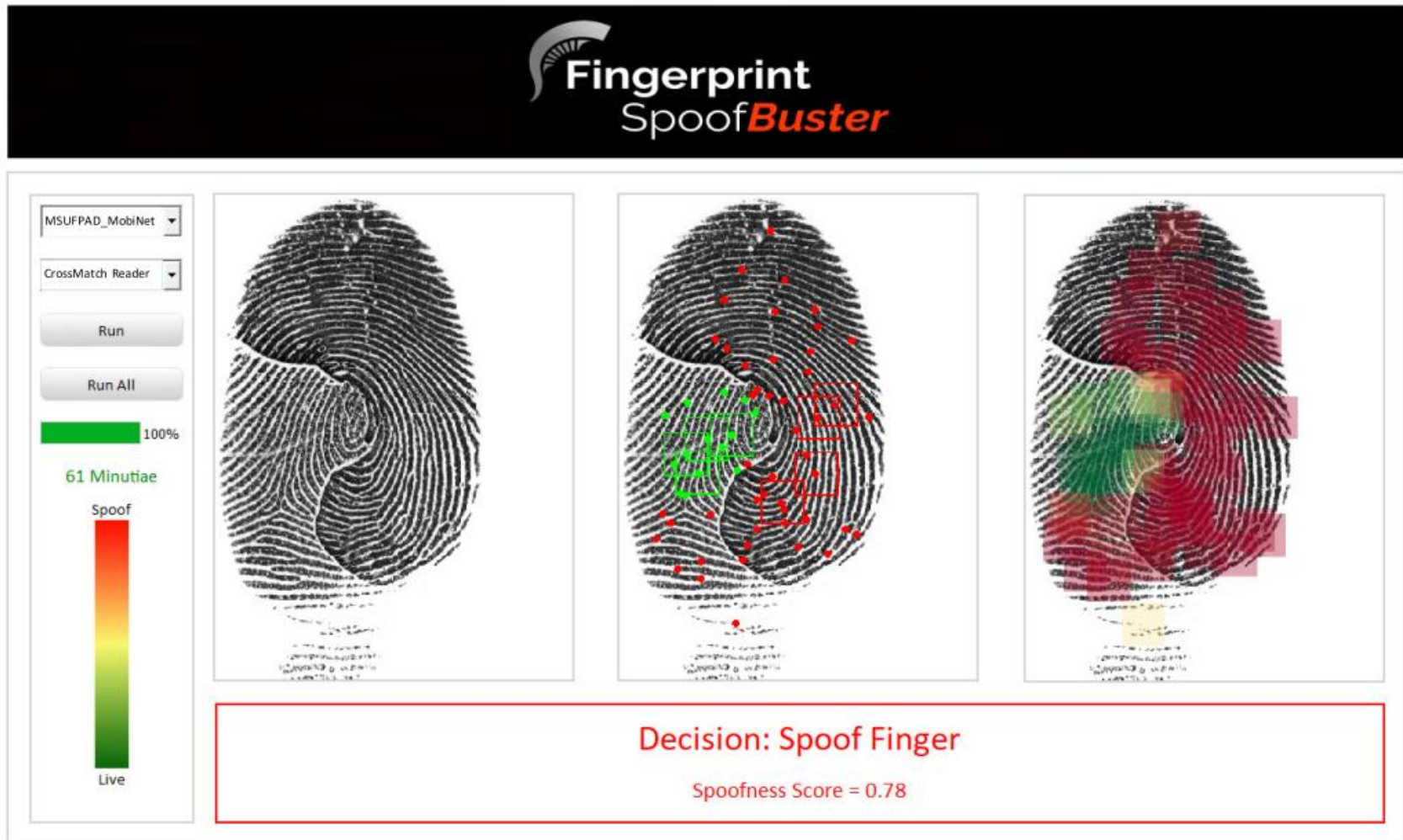


Nogueira et al., Fingerprint Liveness Detection Using Convolutional Neural Networks, IEEE TIFS 2016

Local texture and DNN

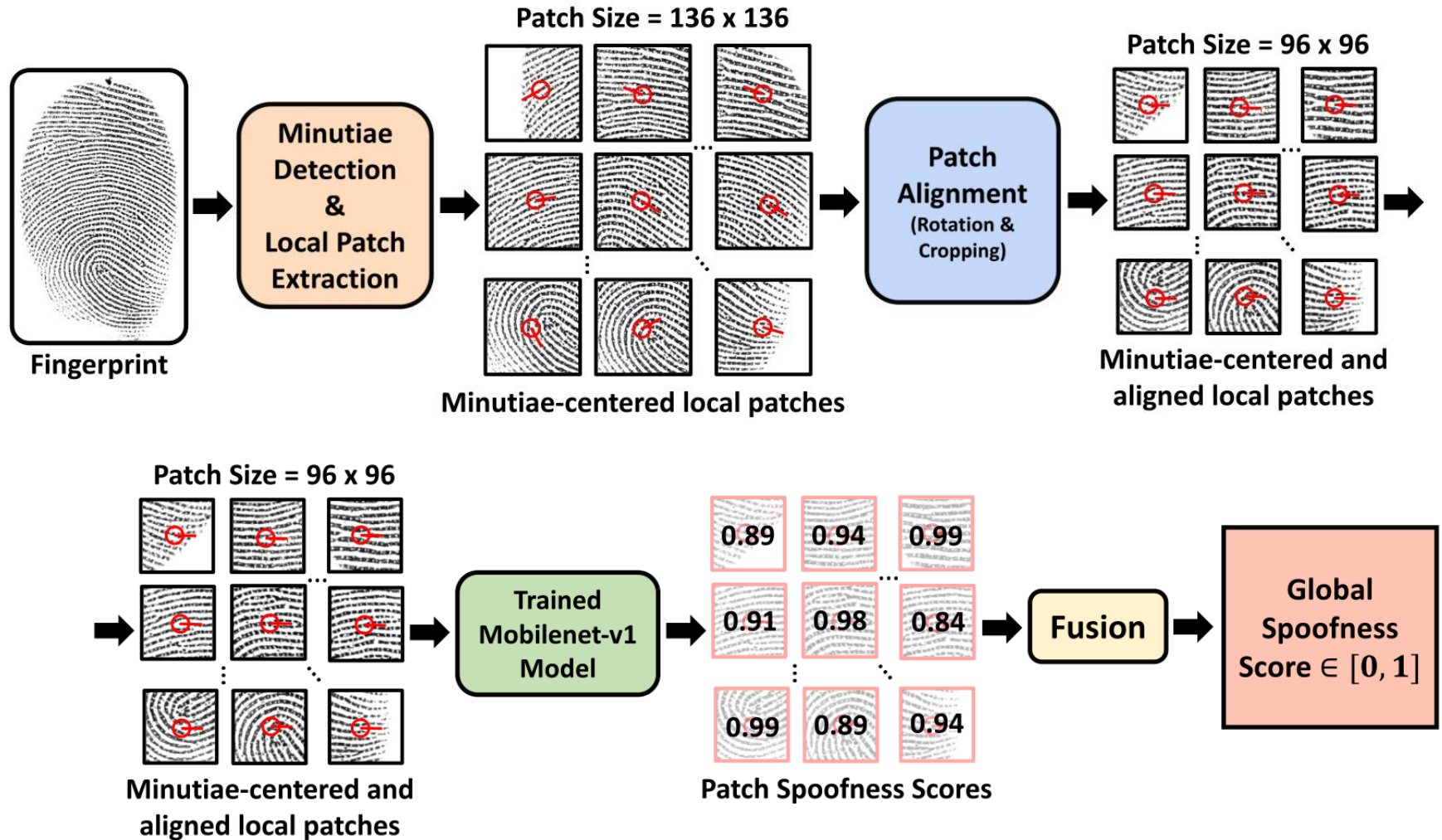


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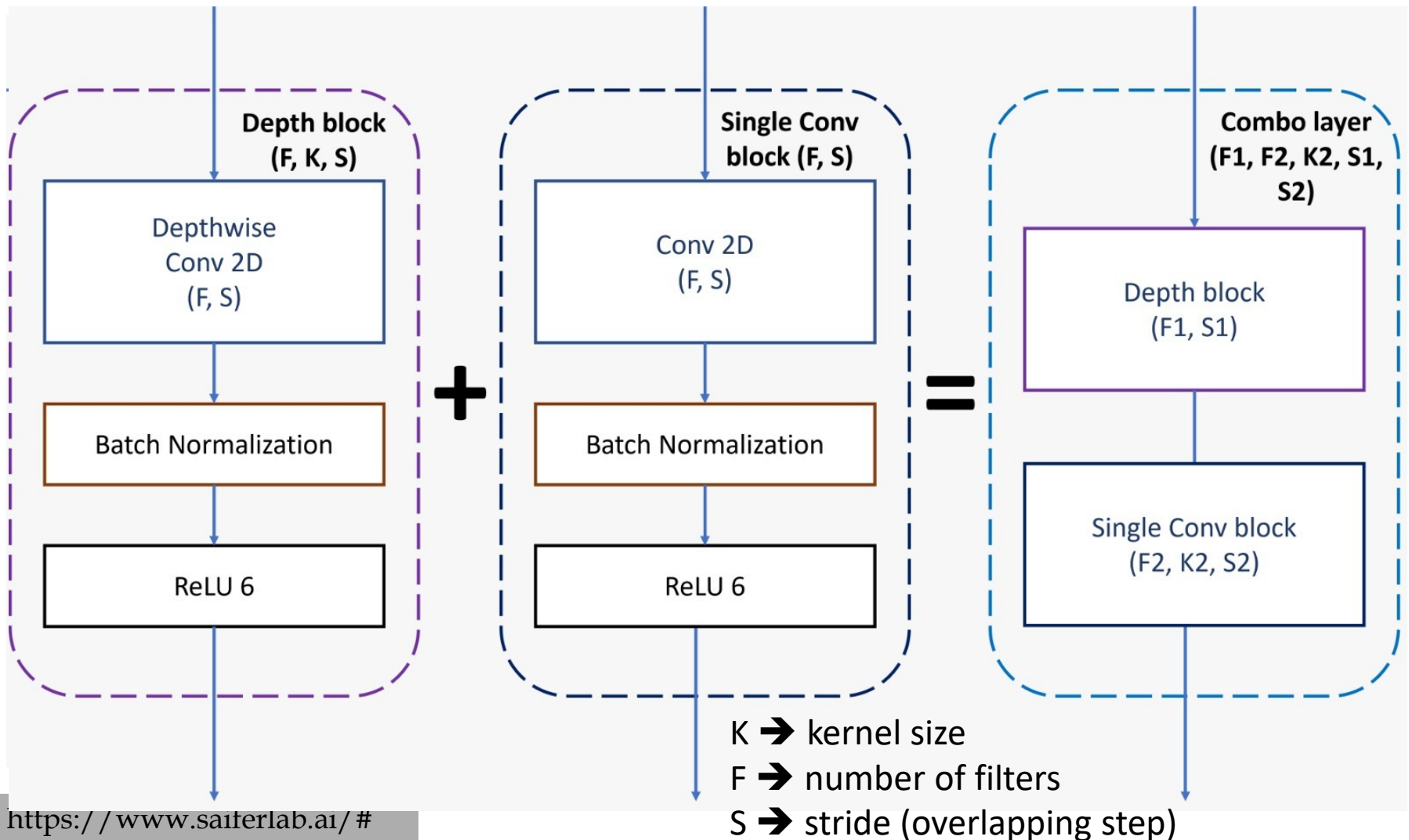
Chug and Jain, Fingerprint Spoof Buster, IEEE TIFS 2018

The Spoof Buster approach



The PRA Lab's Spoof Buster architecture

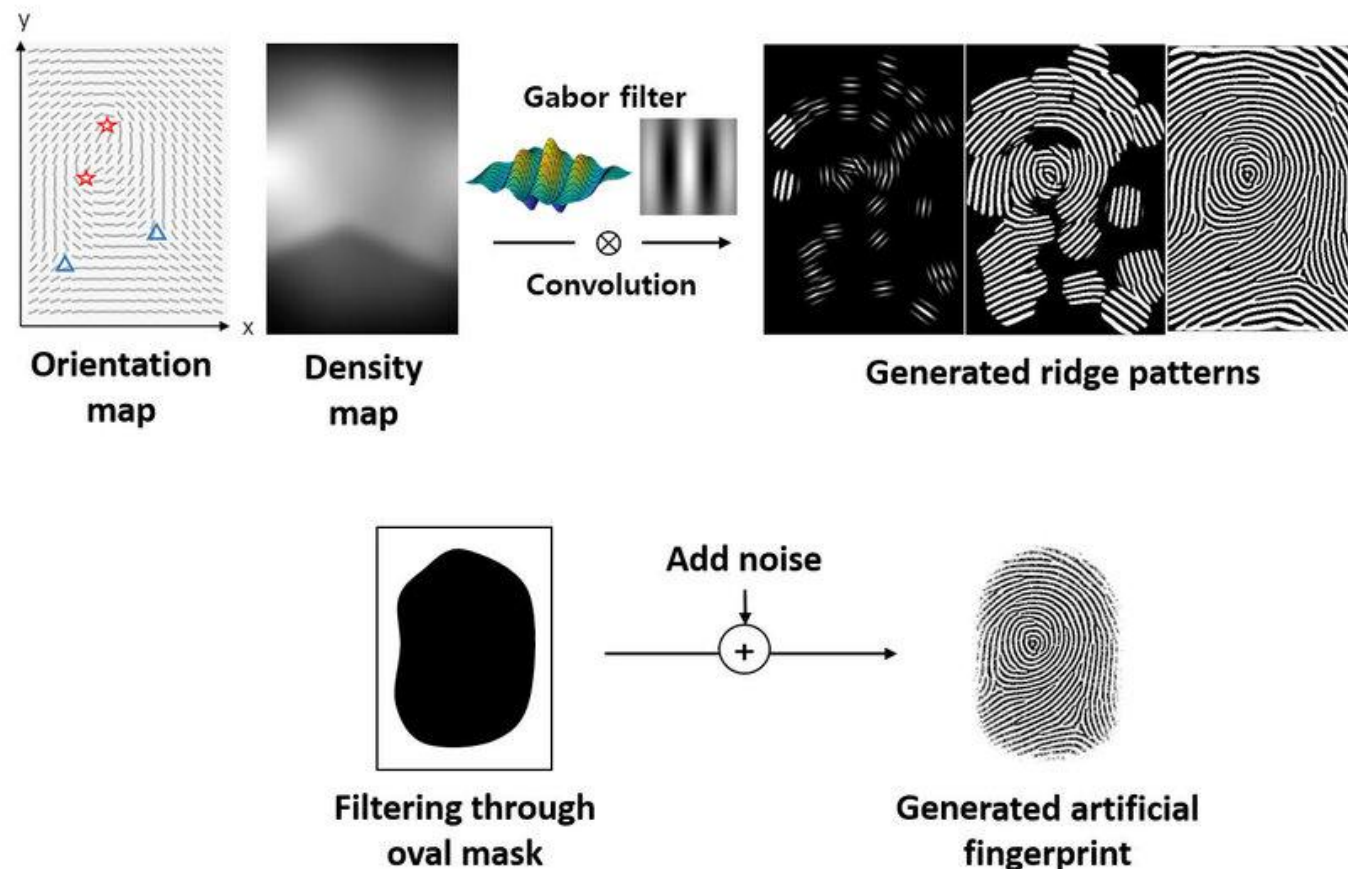
- Inspired from MobileNet



Synthesizing Fingerprints



- Creating an artificial fingerprint image whose appearance is perfectly interchangeable with the one of a «real» fingerprint



D. Hong, *et al.*, FinSNet: End-to-End Separation of Overlapped Fingerprints Using Deep Learning, *IEEE Access*, 8:209020-209029, DOI: 10.1109/ACCESS.2020.3038707, 2020

Why synthetic fingerprint images?



- Evaluating deep-learning systems aimed at substituting minutiae-based identification systems

Measure	DB-1 (Real)		DB-2 (PrintsGAN)	
	Mean	Std. Dev.	Mean	Std. Dev.
Total Minutiae Count	92.70	24.16	79.30	16.59
Ridge Ending Minutiae Count	49.98	16.06	43.00	10.14
Ridge Bifurcation Minutiae Count	42.71	13.36	36.30	9.37
Quality of Minutiae	73.50	15.07	72.14	16.05
Fingerprint Area (Megapixels)	0.179	0.038	0.171	0.019
Fingerprint Image Quality (NFIQ2)	54.21	22.73	63.63	21.38

- Reducing the impact of samples lack for fingerprint PAD based on deep learning

FAR	LivDet	SpoofGAN
0.01% @ threshold=48	99.87%	100%
0.001% @ threshold=60	99.80%	100%
0.0001% @ threshold=72	99.74%	99.80%
1e-05% @ threshold=84	99.41%	99.47%
1e-06% @ threshold=96	99.35%	98.87%

Example of synthetic fingerprint images



(a)



(c)

Live-style fingerprint images



(b)



(d)

Spoof-style fingerprint images

S.A. Grosz and A.K. Jain, SpoofGAN: Synthetic Fingerprint Spoof Images, *IEEE TIFS*, doi: 10.1109/TIFS.2022.3227762, 2023

LivDet competitions



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DERMALOG

THALES



Eight editions (nineth one to come)

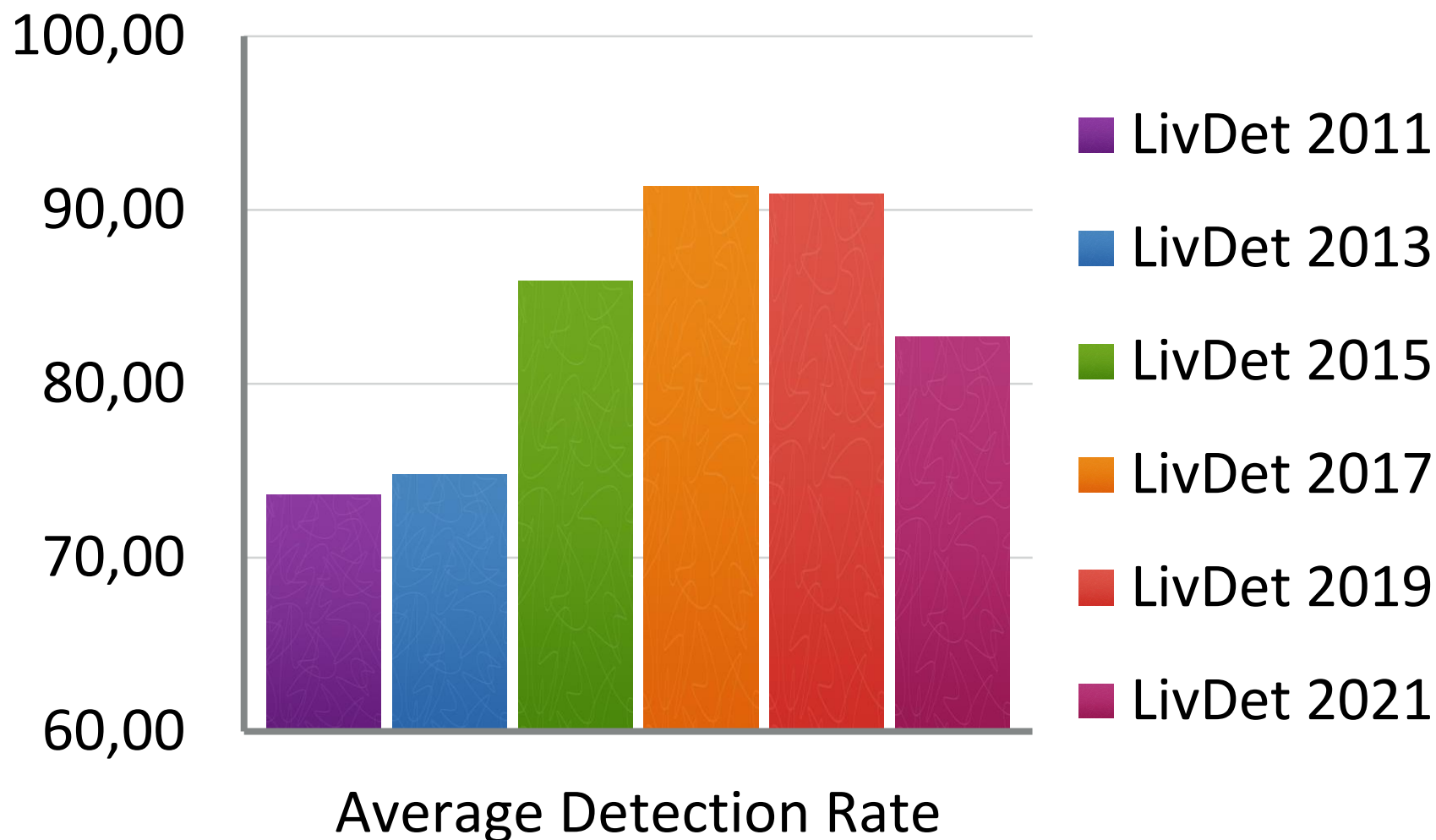


- **Each competition provided up to 4 data sets**
 - 4,000 images per data set
 - Different user population
 - Materials
 - Challenges
- **Each participant sent its fingerprint liveness detection system according to the competition's constraints**
- **We computed the performance of each system according to never-seen-before samples**

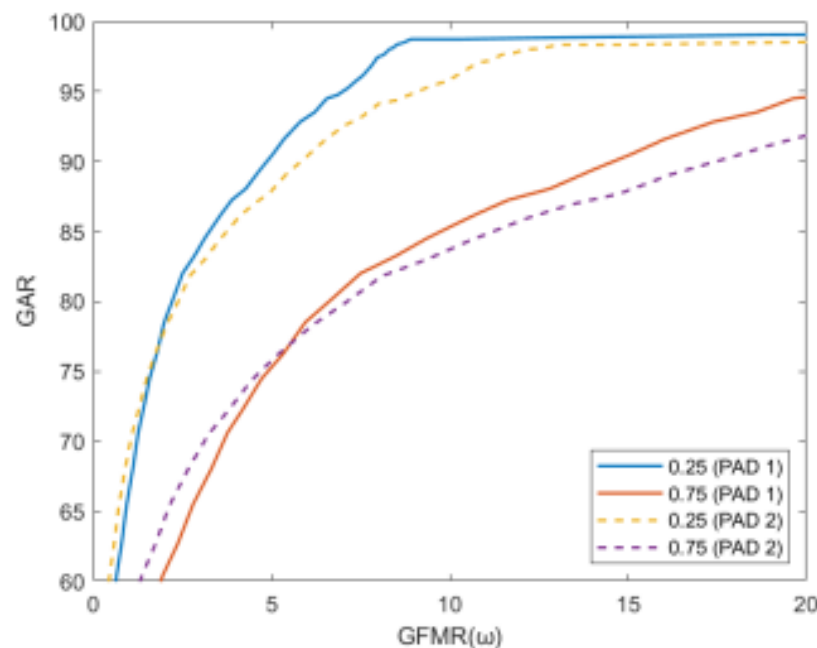
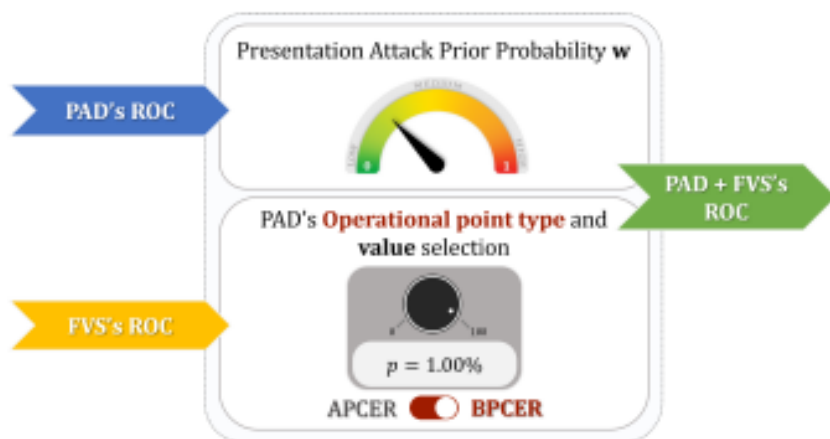
Current state



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Combining PAD and verification systems



Global ROC (GROC) of top 2 PADs submitted to LivDet 2019, integrated with the top off-the-shelf verification system (Verifinger)

M = probability of match
 F = probability of authentic bio

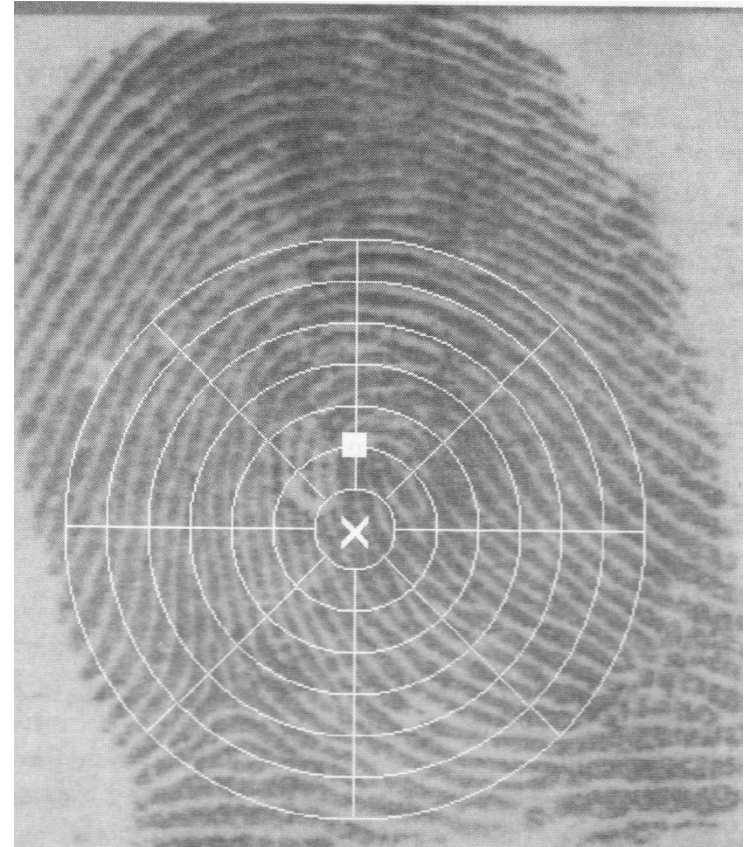
$$GROC = \begin{cases} GAR(M, F) \\ GFMR(M, F; \omega) = FMR(M, F) \cdot (1 - \omega) + IAPMR(M, F) \cdot \omega \end{cases}$$

M. Micheletto, G.L. Marcialis, G. Orrù, F. Roli, Fingerprint recognition with embedded presentation attacks detection: are we ready?, *IEEE Transactions on Image Forensics and Security*, 16 5338-5351, DOI: [10.1109/TIFS.2021.3121201](https://doi.org/10.1109/TIFS.2021.3121201), 2021.

Homework



- **What about feature extraction based on Gabor's filters?**
 - Try the *finger-code*
- **Template-based or machine learning based?**
- **Deep learning models and approaches**
- **Could you provide some home-made spoof?**
 - Gelatine, woodglue, plasticine, das are wellcome!



That's all for today... what's next?



- **Face recognition systems**

- properties
- pros and cons

- **Face liveness detection**

- The problem
- State of the art
- Solutions

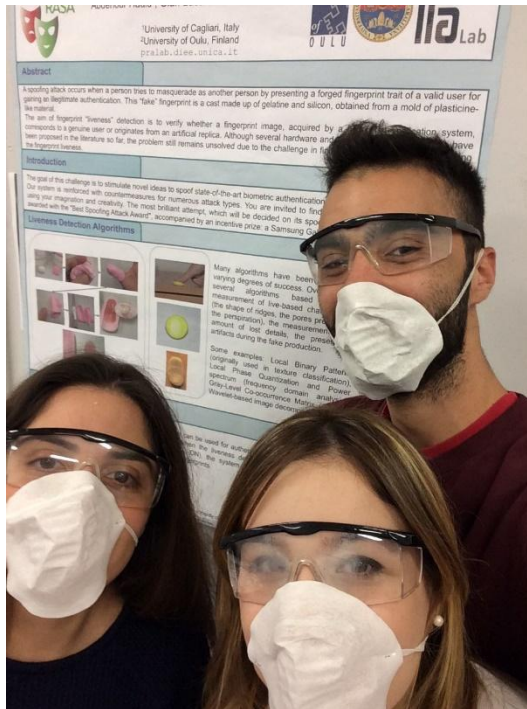
- **Deep fake-faces**

- A modern challenge





Thank you for listening!



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